

1. Source Code

CreatorIQ has been modified into an API-ready architecture consisting of an interactive HTML5 and Tailwind CSS frontend client with native Drag & Drop, File API, video preview player, and a Python Flask REST API backend prepared for future FFmpeg/FFprobe media extraction, YouTube Data and Analytics APIs, and machine learning inference pipelines.

1.1 Frontend Interface & Fetch API Client - index.html

```
<!DOCTYPE html>
<html lang="en" class="scroll-smooth">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CreatorIQ -- AI-Powered Pre-Upload YouTube Analytics</title>

  <!-- Google Fonts: Plus Jakarta Sans -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@300;400;500;600;700;800&display=swap" rel="stylesheet">

  <!-- Tailwind CSS CDN -->
  <script src="https://cdn.tailwindcss.com"></script>

  <!-- Custom Styles & Clean Keyframe Animations -->
  <style>
    body {
      font-family: 'Plus Jakarta Sans', sans-serif;
    }

    /* Subtle grid background pattern */
    .bg-grid-pattern {
      background-size: 40px 40px;
      background-image:
        linear-gradient(to right, rgba(255, 255, 255, 0.03) 1px, transparent 1px),
        linear-gradient(to bottom, rgba(255, 255, 255, 0.03) 1px, transparent 1px);
    }

    /* Gentle floating animations */
    @keyframes float {
      0%, 100% { transform: translateY(0px); }
      50% { transform: translateY(-10px); }
    }

    .animate-float {
      animation: float 5s ease-in-out infinite;
    }

    .animate-float-delayed {
      animation: float 6s ease-in-out 1s infinite;
    }
  </style>
</head>

<body class="bg-black text-zinc-100 antialiased selection:bg-red-600 selection:text-white min-h-screen relative overflow-x-hidden">

  <!-- Ambient Red Glow Background Effects -->
  <div class="fixed inset-0 pointer-events-none z-0 overflow-hidden">
    <div class="absolute -top-32 left-1/2 -translate-x-1/2 w-[600px] h-[450px] bg-red-600 rounded-full blur-[140px] opacity-20"></div>
    <div class="absolute top-[40%] -right-32 w-[500px] h-[500px] bg-red-700 rounded-full blur-[150px] opacity-15"></div>
    <div class="absolute bottom-20 -left-32 w-[500px] h-[500px] bg-red-600 rounded-full blur-[140px] opacity-15"></div>
  </div>

  <!-- Background Grid Pattern -->
  <div class="fixed inset-0 bg-grid-pattern opacity-60 pointer-events-none z-0"></div>

  <!-- ===== -->
  <!-- 1. NAVBAR (Sticky Glassmorphic Header) -->
  <!-- ===== -->
  <header class="sticky top-0 z-50 w-full border-b border-zinc-800/80 bg-black/80 backdrop-blur-xl transition-all">
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
      <div class="flex items-center justify-between h-20">

        <!-- Logo & Brand Name -->
        <a href="#" class="flex items-center gap-3 group">
          <div class="w-10 h-10 rounded-xl bg-gradient-to-tr from-red-600 to-rose-500 flex items-center justify-center shadow">
            <svg class="w-5 h-5 text-white fill-current" viewBox="0 0 24 24">
              <path d="M8 5v14l11-7z"/>
            </svg>
          </div>
          <div class="flex flex-col">
            <span class="text-xl font-extrabold tracking-tight text-white flex items-center gap-1">
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        Creator<span class="text-transparent bg-clip-text bg-gradient-to-r from-red-500 to-rose-400">IQ</span>
      </span>
      <span class="text-[10px] tracking-widest uppercase text-zinc-400 font-semibold -mt-1">Pre-Upload AI</span>
    </div>
  </a>

<!-- Desktop Navigation Links -->
<nav class="hidden md:flex items-center space-x-2">
  <a href="#features" class="px-4 py-2 text-sm font-medium text-zinc-300 hover:text-white hover:bg-zinc-900/60 rounded-lg">Features</a>
  <a href="#how-it-works" class="px-4 py-2 text-sm font-medium text-zinc-300 hover:text-white hover:bg-zinc-900/60 rounded-lg">How it works</a>
  <a href="#about" class="px-4 py-2 text-sm font-medium text-zinc-300 hover:text-white hover:bg-zinc-900/60 rounded-lg">About</a>
</nav>

<!-- Desktop CTA Button -->
<div class="hidden md:flex items-center gap-4">
  <a href="#home" class="inline-flex items-center gap-2 px-6 py-2.5 rounded-xl font-semibold text-sm text-white bg-gradient-to-r from-white via-zinc-200 to-zinc-400">
    <span>GET STARTED</span>
    <svg class="w-4 h-4" fill="none" stroke="currentColor" viewBox="0 0 24 24">
      <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M14 5l7 7m0 0l-7 7m7-7H3"/>
    </svg>
  </a>
</div>

<!-- Pure CSS Mobile Menu Toggle -->
<input type="checkbox" id="nav-toggle" class="hidden peer">
<label for="nav-toggle" class="md:hidden p-2.5 rounded-xl text-zinc-400 hover:text-white hover:bg-zinc-900 border border-zinc-300">
  <svg class="w-6 h-6" fill="none" stroke="currentColor" viewBox="0 0 24 24">
    <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M4 6h16M4 12h16M4 18h16"/>
  </svg>
</label>

<!-- Mobile Dropdown Drawer -->
<div class="peer-checked:flex hidden flex-col absolute top-20 left-0 w-full bg-zinc-950/95 backdrop-blur-2xl border-b border-zinc-800">
  <a href="#features" class="text-zinc-300 hover:text-white text-base font-medium py-2 border-b border-zinc-900">Features</a>
  <a href="#how-it-works" class="text-zinc-300 hover:text-white text-base font-medium py-2 border-b border-zinc-900">How it works</a>
  <a href="#about" class="text-zinc-300 hover:text-white text-base font-medium py-2 border-b border-zinc-900">About</a>
  <a href="#home" class="flex items-center justify-center gap-2 w-full py-3 rounded-xl font-semibold text-white bg-gradient-to-r from-white via-zinc-200 to-zinc-400">
    <span>GET STARTED</span>
    <svg class="w-4 h-4" fill="none" stroke="currentColor" viewBox="0 0 24 24">
      <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M14 5l7 7m0 0l-7 7m7-7H3"/>
    </svg>
  </a>
</div>
</div>
</div>
</header>

<!-- ===== -->
<!-- 2. HERO SECTION -->
<!-- ===== -->
<section class="relative z-10 pt-12 pb-20 md:pt-20 md:pb-32 overflow-hidden">
  <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
    <div class="grid grid-cols-1 lg:grid-cols-12 gap-12 lg:gap-8 items-center">

      <!-- Left Column: Copy & Actions -->
      <div class="lg:col-span-6 space-y-8 text-center lg:text-left">

        <!-- Red Badge -->
        <div class="inline-flex items-center gap-2 px-3.5 py-1.5 rounded-full bg-red-950/60 border border-red-800/60 text-white">
          <span class="w-2 h-2 rounded-full bg-red-500 animate-ping"></span>
          <span>AI-POWERED YOUTUBE ANALYTICS</span>
        </div>

        <!-- Heading -->
        <h1 class="text-4xl sm:text-5xl lg:text-6xl font-extrabold tracking-tight text-white leading-tight">
          Make Every <br class="hidden sm:block">
          <span class="text-transparent bg-clip-text bg-gradient-to-r from-white via-zinc-200 to-zinc-400">
            Upload Count.
          </span>
        </h1>

        <!-- Subtitle -->
        <p class="text-base sm:text-lg text-zinc-400 max-w-xl mx-auto lg:mx-0 leading-relaxed">
          Analyze your video before publishing. Discover its quality, engagement potential, and what you can improve before

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</p>

<!-- CTAs -->
<div class="pt-2 flex flex-col sm:flex-row items-center justify-center lg:justify-start gap-4">
  <a href="#home" class="w-full sm:w-auto inline-flex items-center justify-center gap-3 px-8 py-4 rounded-xl font-medium">
    <span>Get Started -></span>
  </a>
  <a href="#features" class="w-full sm:w-auto inline-flex items-center justify-center px-6 py-4 rounded-xl font-medium">
    Explore Features
  </a>
</div>

<!-- Tagline -->
<div class="pt-2 flex items-center justify-center lg:justify-start gap-3 text-xs tracking-widest uppercase font-medium">
  <span class="text-red-400">Analyze</span>
  <span class="text-zinc-600">*</span>
  <span class="text-zinc-200">Improve</span>
  <span class="text-zinc-600">*</span>
  <span class="text-red-400">Upload</span>
</div>

<!-- Trust Stats Bar -->
<div class="pt-6 border-t border-zinc-900 grid grid-cols-3 gap-4 max-w-md mx-auto lg:mx-0 text-left">
  <div>
    <div class="text-xl font-bold text-white">99.4%</div>
    <div class="text-[11px] text-zinc-400">Accuracy Rate</div>
  </div>
  <div>
    <div class="text-xl font-bold text-white">&lt; 30s</div>
    <div class="text-[11px] text-zinc-400">Full Analysis</div>
  </div>
  <div>
    <div class="text-xl font-bold text-white">4K 60fps</div>
    <div class="text-[11px] text-zinc-400">HDR Support</div>
  </div>
</div>
</div>

<!-- Right Column: Interactive Dashboard Mockup -->
<div class="lg:col-span-6 relative flex items-center justify-center py-6">

  <!-- Ambient Glow -->
  <div class="absolute inset-0 bg-red-600/20 blur-3xl rounded-3xl pointer-events-none"></div>

  <!-- Main Floating Dashboard Card -->
  <div class="relative w-full max-w-lg bg-zinc-950/90 backdrop-blur-2xl border border-zinc-800 rounded-2xl p-6 sm:p-7">

    <!-- Header -->
    <div class="flex items-center justify-between pb-5 border-b border-zinc-800">
      <div class="flex items-center gap-3">
        <div class="w-8 h-8 rounded-lg bg-red-600/20 border border-red-500/30 flex items-center justify-center">
          <svg class="w-4 h-4" fill="none" stroke="currentColor" viewBox="0 0 24 24">
            <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M15 10l4.53 2.276A1 1 0 0 0 17 10l-4.53-2.276A1 1 0 0 0 15 10z"/>
          </svg>
        </div>
        <div>
          <div class="text-xs font-bold uppercase tracking-wider text-red-500">CreatorIQ Engine</div>
          <div class="text-sm font-semibold text-white">Video Analysis</div>
        </div>
      </div>
      <div class="inline-flex items-center gap-1.5 px-2.5 py-1 rounded-full bg-emerald-950/50 border border-emerald-400">
        <span class="w-1.5 h-1.5 rounded-full bg-emerald-400"></span>
        Scanned
      </div>
    </div>

    <!-- Score Progress -->
    <div class="mt-6 p-4 rounded-xl bg-zinc-900/60 border border-zinc-800">
      <div class="flex items-baseline justify-between mb-2">
        <span class="text-xs font-bold tracking-wider uppercase text-zinc-400">VIDEO QUALITY</span>
        <div class="flex items-baseline gap-1">
          <span class="text-3xl font-extrabold text-white">87</span>
          <span class="text-sm font-semibold text-zinc-400">/ 100</span>
        </div>
      </div>
      <div class="w-full bg-zinc-800 rounded-full h-2.5 overflow-hidden">

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<div class="bg-gradient-to-r from-red-600 to-rose-500 h-2.5 rounded-full w-[87%]"></div>
</div>
<div class="flex justify-between items-center mt-2 text-[11px] text-zinc-400">
  <span>Optimal YouTube standard met</span>
  <span class="text-emerald-400 font-semibold">Top 12% in Category</span>
</div>
</div>
<!-- Metrics Breakdown -->
<div class="mt-5 space-y-2.5">
  <div class="flex items-center justify-between p-3 rounded-lg bg-zinc-900/40 border border-zinc-800 hover:bor
    <div class="flex items-center gap-2.5">
      <span class="text-sm">[VIDEO]</span>
      <span class="text-xs font-medium text-zinc-300">Resolution</span>
    </div>
    <span class="px-2.5 py-0.5 rounded-md bg-emerald-950/60 border border-emerald-600/40 text-emerald-400 t
  </div>

  <div class="flex items-center justify-between p-3 rounded-lg bg-zinc-900/40 border border-zinc-800 hover:bor
    <div class="flex items-center gap-2.5">
      <span class="text-sm">[AUDIO]</span>
      <span class="text-xs font-medium text-zinc-300">Audio</span>
    </div>
    <span class="px-2.5 py-0.5 rounded-md bg-zinc-800 border border-zinc-700 text-zinc-300 text-xs font-sem
  </div>

  <div class="flex items-center justify-between p-3 rounded-lg bg-zinc-900/40 border border-zinc-800 hover:bor
    <div class="flex items-center gap-2.5">
      <span class="text-sm">[LIGHT]</span>
      <span class="text-xs font-medium text-zinc-300">Lighting</span>
    </div>
    <span class="px-2.5 py-0.5 rounded-md bg-zinc-800 border border-zinc-700 text-zinc-300 text-xs font-sem
  </div>

  <div class="flex items-center justify-between p-3 rounded-lg bg-zinc-900/40 border border-zinc-800 hover:bor
    <div class="flex items-center gap-2.5">
      <span class="text-sm">[SEARCH]</span>
      <span class="text-xs font-medium text-zinc-300">Sharpness</span>
    </div>
    <span class="px-2.5 py-0.5 rounded-md bg-emerald-950/60 border border-emerald-600/40 text-emerald-400 t
  </div>
</div>
<!-- Card Footer -->
<div class="mt-5 pt-4 border-t border-zinc-800 flex items-center justify-between text-xs text-zinc-400">
  <span class="flex items-center gap-1.5">
    <span class="w-2 h-2 rounded-full bg-red-500"></span>
    Analyzed with Neural Engine 3.2
  </span>
  <span class="text-zinc-300 font-mono">sample_vlog_final.mp4</span>
</div>
</div>
<!-- Satellite Badge 1: Top Right (Quality Score) -->
<div class="absolute -top-4 sm:-top-6 -right-2 sm:-right-6 bg-zinc-900/90 backdrop-blur-md border border-red-500/40
  <div class="flex items-center gap-2.5">
    <div class="w-8 h-8 rounded-lg bg-red-600/20 text-red-400 flex items-center justify-center font-bold text-s
    <div>
      <div class="text-[11px] text-zinc-400 uppercase tracking-wider font-semibold">Score</div>
      <div class="text-xs font-bold text-white">87% Quality</div>
    </div>
  </div>
</div>
<!-- Satellite Badge 2: Bottom Left (High Engagement) -->
<div class="absolute -bottom-6 -left-3 sm:-left-8 bg-zinc-900/90 backdrop-blur-md border border-zinc-700 rounded-xl
  <div class="flex items-center gap-3">
    <div class="w-9 h-9 rounded-lg bg-emerald-500/20 text-emerald-400 flex items-center justify-center">
      <svg class="w-5 h-5" fill="none" stroke="currentColor" viewBox="0 0 24 24">
        <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M13 7h8m0 0v8m0-8l-8 8-4-4"
      </svg>
    </div>
    <div>
      <div class="text-[11px] text-zinc-400 uppercase tracking-wider font-semibold">Prediction</div>
      <div class="text-xs font-bold text-emerald-400 flex items-center gap-1">
        <span>High Engagement</span>
      </div>
    </div>
  </div>
</div>
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        <span>[HOT]</span>
      </div>
    </div>
  </div>
</div>

<!-- Satellite Badge 3: Bottom Right (AI Recommendation) -->
<div class="hidden sm:flex absolute -bottom-5 -right-6 bg-zinc-900/90 backdrop-blur-md border border-red-500/30 rounded-2xl">
  <div class="flex items-center gap-2.5">
    <div class="w-8 h-8 rounded-lg bg-red-600/20 text-red-400 flex items-center justify-center">[AI]</div>
    <div>
      <div class="text-[11px] text-zinc-400 uppercase tracking-wider font-semibold">AI Assistant</div>
      <div class="text-xs font-bold text-white">AI Recommendation</div>
    </div>
  </div>
</div>

</div>
</div>
</section>

<!-- ===== -->
<!-- 3. FEATURES SECTION -->
<!-- ===== -->
<section id="features" class="relative z-10 py-20 border-t border-zinc-900 bg-gradient-to-b from-transparent via-zinc-950/40 to-transparent">
  <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">

    <!-- Section Header -->
    <div class="text-center max-w-3xl mx-auto mb-16 space-y-4">
      <div class="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-red-950/40 border border-red-900/50 text-red-400">
        Core Capabilities
      </div>
      <h2 class="text-3xl sm:text-4xl font-extrabold text-white tracking-tight">
        Everything You Need Before You Upload
      </h2>
      <p class="text-base text-zinc-400 leading-relaxed">
        Stop guessing how the YouTube algorithm and viewers will react. Validate your production before making it public.
      </p>
    </div>

    <!-- 3 Feature Cards Grid -->
    <div class="grid grid-cols-1 md:grid-cols-3 gap-8">

      <!-- Feature 1: Video Quality -->
      <div class="group rounded-2xl bg-zinc-900/50 hover:bg-zinc-900/80 border border-zinc-800 hover:border-red-500/60 p-8 transition-colors">
        <div class="w-14 h-14 rounded-xl bg-zinc-800/80 group-hover:bg-red-600/20 border border-zinc-700/50 group-hover:border-red-500/60 flex items-center justify-center">
          ■
        </div>
        <h3 class="text-xl font-bold text-white mb-3 group-hover:text-red-400 transition-colors">
          Video Quality
        </h3>
        <p class="text-sm text-zinc-400 leading-relaxed">
          Check resolution, clarity, lighting and audio quality before publishing. Detect compression artifacts, noise and
        </p>
        <div class="mt-6 pt-4 border-t border-zinc-800/60 flex items-center text-xs font-semibold text-zinc-400 group-hover:text-red-400">
          <span>Automated frame scan</span>
          <span class="ml-auto"></span>
        </div>
      </div>

      <!-- Feature 2: Engagement Prediction -->
      <div class="group rounded-2xl bg-zinc-900/50 hover:bg-zinc-900/80 border border-zinc-800 hover:border-red-500/60 p-8 transition-colors">
        <div class="w-14 h-14 rounded-xl bg-zinc-800/80 group-hover:bg-red-600/20 border border-zinc-700/50 group-hover:border-red-500/60 flex items-center justify-center">
          ■
        </div>
        <h3 class="text-xl font-bold text-white mb-3 group-hover:text-red-400 transition-colors">
          Engagement Prediction
        </h3>
        <p class="text-sm text-zinc-400 leading-relaxed">
          Estimate potential views, likes and shares using AI-powered analysis trained on millions of high-performing YouTubers.
        </p>
        <div class="mt-6 pt-4 border-t border-zinc-800/60 flex items-center text-xs font-semibold text-zinc-400 group-hover:text-red-400">
          <span>Pacing & retention modeling</span>
          <span class="ml-auto"></span>
        </div>
      </div>

    </div>

  </div>
</section>
```

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        <span class="ml-auto">-></span>
      </div>
    </div>

    <!-- Feature 3: AI Recommendations -->
    <div class="group rounded-2xl bg-zinc-900/50 hover:bg-zinc-900/80 border border-zinc-800 hover:border-red-500/60 p-8 tra
      <div class="w-14 h-14 rounded-xl bg-zinc-800/80 group-hover:bg-red-600/20 border border-zinc-700/50 group-hover:bor
        [AI]
      </div>
      <h3 class="text-xl font-bold text-white mb-3 group-hover:text-red-400 transition-colors">
        AI Recommendations
      </h3>
      <p class="text-sm text-zinc-400 leading-relaxed">
        Discover changes that could improve your video's performance. Get actionable advice on hook pacing, audio duckin
      </p>
      <div class="mt-6 pt-4 border-t border-zinc-800/60 flex items-center text-xs font-semibold text-zinc-400 group-hover:
        <span>Actionable checklist</span>
        <span class="ml-auto">-></span>
      </div>
    </div>

  </div>
</div>
</section>

<!-- ===== -->
<!-- 4. HOW IT WORKS SECTION -->
<!-- ===== -->
<section id="how-it-works" class="relative z-10 py-20 border-t border-zinc-900">
  <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">

    <!-- Section Title -->
    <div class="text-center max-w-2xl mx-auto mb-16 space-y-3">
      <div class="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-red-950/40 border border-red-900/50 text-red-400 t
        Simple 4-Step Workflow
      </div>
      <h2 class="text-3xl sm:text-4xl font-extrabold text-white tracking-tight">
        From Edit to Upload
      </h2>
      <p class="text-sm sm:text-base text-zinc-400">
        Integrate CreatorIQ into your export pipeline without adding friction.
      </p>
    </div>

    <!-- Steps Grid (4 Columns) -->
    <div class="grid grid-cols-1 md:grid-cols-4 gap-6 relative">

      <!-- Step 01 -->
      <div class="relative flex flex-col p-6 rounded-2xl bg-zinc-900/40 border border-zinc-800 hover:border-red-500/40 transi
        <div class="flex items-center justify-between mb-4">
          <span class="text-3xl font-black text-transparent bg-clip-text bg-gradient-to-r from-red-500 to-rose-400">01</sp
          <div class="w-8 h-8 rounded-lg bg-zinc-800 flex items-center justify-center text-xs text-zinc-400">[FILE]</div>
        </div>
        <h3 class="text-lg font-bold uppercase tracking-wider text-white mb-2">UPLOAD</h3>
        <p class="text-sm text-zinc-400 leading-relaxed">Upload your edited video.</p>
        <div class="mt-4 text-[11px] text-zinc-400">Supports MP4, MOV, MKV</div>
        <div class="hidden md:block absolute -right-3 top-1/2 -translate-y-1/2 text-zinc-600 z-20 font-mono text-sm">-></div>
        <div class="md:hidden text-center text-red-500/60 my-2">↓</div>
      </div>

      <!-- Step 02 -->
      <div class="relative flex flex-col p-6 rounded-2xl bg-zinc-900/40 border border-zinc-800 hover:border-red-500/40 transi
        <div class="flex items-center justify-between mb-4">
          <span class="text-3xl font-black text-transparent bg-clip-text bg-gradient-to-r from-red-500 to-rose-400">02</sp
          <div class="w-8 h-8 rounded-lg bg-zinc-800 flex items-center justify-center text-xs text-zinc-400">[FAST]</div>
        </div>
        <h3 class="text-lg font-bold uppercase tracking-wider text-white mb-2">ANALYZE</h3>
        <p class="text-sm text-zinc-400 leading-relaxed">AI evaluates the video.</p>
        <div class="mt-4 text-[11px] text-zinc-400">Frames, audio & pacing</div>
        <div class="hidden md:block absolute -right-3 top-1/2 -translate-y-1/2 text-zinc-600 z-20 font-mono text-sm">-></div>
        <div class="md:hidden text-center text-red-500/60 my-2">↓</div>
      </div>

      <!-- Step 03 -->
      <div class="relative flex flex-col p-6 rounded-2xl bg-zinc-900/40 border border-zinc-800 hover:border-red-500/40 transi

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<div class="flex items-center justify-between mb-4">
  <span class="text-3xl font-black text-transparent bg-clip-text bg-gradient-to-r from-red-500 to-rose-400">03</span>
</div>
<h3 class="text-lg font-bold uppercase tracking-wider text-white mb-2">IMPROVE</h3>
<p class="text-sm text-zinc-400 leading-relaxed">Follow the recommendations.</p>
<div class="mt-4 text-[11px] text-zinc-400">Apply AI suggested tweaks</div>
<div class="hidden md:block absolute -right-3 top-1/2 -translate-y-1/2 text-zinc-600 z-20 font-mono text-sm">-</div>
<div class="md:hidden text-center text-red-500/60 my-2">↓</div>
</div>

<!-- Step 04 -->
<div class="relative flex flex-col p-6 rounded-2xl bg-zinc-900/40 border border-zinc-800 hover:border-red-500/40 transition">
  <div class="flex items-center justify-between mb-4">
    <span class="text-3xl font-black text-transparent bg-clip-text bg-gradient-to-r from-red-500 to-rose-400">04</span>
    <div class="w-8 h-8 rounded-lg bg-zinc-800 flex items-center justify-center text-xs text-zinc-400">[ROCKET]</div>
  </div>
  <h3 class="text-lg font-bold uppercase tracking-wider text-white mb-2">PUBLISH</h3>
  <p class="text-sm text-zinc-400 leading-relaxed">Upload your optimized video.</p>
  <div class="mt-4 text-[11px] text-zinc-400">Ship with total confidence</div>
</div>

</div>
</div>
</section>

<!-- ===== -->
<!-- 5. ABOUT SECTION -->
<!-- ===== -->
<section id="about" class="relative z-10 py-20 border-t border-zinc-900 bg-zinc-950/60">
  <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
    <div class="grid grid-cols-1 lg:grid-cols-12 gap-12 items-center">

      <!-- Left: Story & Stats -->
      <div class="lg:col-span-7 space-y-6">
        <div class="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-red-950/40 border border-red-900/50 text-red-400">
          Why CreatorIQ
        </div>
        <h2 class="text-3xl sm:text-4xl font-extrabold text-white tracking-tight">
          Engineered for High-Growth YouTube Creators
        </h2>
        <p class="text-base text-zinc-400 leading-relaxed">
          Every day, creators spend dozens of hours editing only to have YouTube's initial impressions flatline due to av
        </p>
        <p class="text-base text-zinc-400 leading-relaxed">
          CreatorIQ acts as your objective AI copilot. By evaluating frame dynamics, audio fidelity, and historical viewe
        </p>

        <!-- Stat Counters -->
        <div class="grid grid-cols-2 sm:grid-cols-3 gap-4 pt-4">
          <div class="p-4 rounded-xl bg-zinc-900/60 border border-zinc-800">
            <div class="text-2xl font-bold text-white">10M+</div>
            <div class="text-xs text-zinc-400 mt-1">Data Points Analyzed</div>
          </div>
          <div class="p-4 rounded-xl bg-zinc-900/60 border border-zinc-800">
            <div class="text-2xl font-bold text-white">0%</div>
            <div class="text-xs text-zinc-400 mt-1">Quality Compromise</div>
          </div>
          <div class="p-4 rounded-xl bg-zinc-900/60 border border-zinc-800 col-span-2 sm:col-span-1">
            <div class="text-2xl font-bold text-red-500">100%</div>
            <div class="text-xs text-zinc-400 mt-1">Creator-Centric</div>
          </div>
        </div>
      </div>

      <!-- Right: Pre-Upload Checklist Card -->
      <div class="lg:col-span-5 relative">
        <div class="p-6 rounded-2xl bg-zinc-900/60 border border-zinc-800 space-y-4 shadow-xl">
          <div class="flex items-center justify-between pb-3 border-b border-zinc-800">
            <span class="text-xs font-bold uppercase text-zinc-400">Pre-Upload Checklist</span>
            <span class="text-xs text-red-400 font-semibold">AI Diagnostic Active</span>
          </div>

          <div class="space-y-2.5 text-xs">
            <div class="p-2.5 rounded-lg bg-zinc-950/80 border border-zinc-800 flex items-center justify-between">

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        <span class="text-zinc-300">Thumbnail / Hook Contrast</span>
        <span class="text-emerald-400 font-bold">Optimized</span>
      </div>
      <div class="p-2.5 rounded-lg bg-zinc-950/80 border border-zinc-800 flex items-center justify-between">
        <span class="text-zinc-300">Loudness Compliance (-14 LUFS)</span>
        <span class="text-emerald-400 font-bold">Passed</span>
      </div>
      <div class="p-2.5 rounded-lg bg-zinc-950/80 border border-zinc-800 flex items-center justify-between">
        <span class="text-zinc-300">Bitrate & Compression artifacts</span>
        <span class="text-emerald-400 font-bold">Clear</span>
      </div>
      <div class="p-2.5 rounded-lg bg-zinc-950/80 border border-zinc-800 flex items-center justify-between">
        <span class="text-zinc-300">Audience Retention Drop Probability</span>
        <span class="text-zinc-300 font-bold">Low (11%)</span>
      </div>
    </div>

    <div class="pt-2 text-center">
      <a href="#home" class="text-xs text-red-400 hover:text-red-300 font-semibold inline-flex items-center gap-1">
        Test Your Video Now <span>-></span>
      </a>
    </div>
  </div>
</div>
</section>

<!-- ===== -->
<!-- 6. HOME SECTION (Workspace & Analytics Preview) -->
<!-- ===== -->
<section id="home" class="relative z-10 py-24 border-t border-zinc-800/80 bg-zinc-950/80">
  <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">

    <!-- Section Title -->
    <div class="text-center max-w-3xl mx-auto mb-16 space-y-4">
      <div class="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-red-950/40 border border-red-900/50 text-red-400">
        CreatorIQ Workspace
      </div>
      <h2 class="text-3xl sm:text-5xl font-extrabold text-white tracking-tight">
        Welcome to CreatorIQ
      </h2>
      <p class="text-base sm:text-lg text-zinc-400">
        Prepare your next YouTube video before you publish.
      </p>
    </div>

    <!-- ===== -->
    <!-- 1. LARGE VIDEO UPLOAD & INTERACTIVE PREVIEW CARD -->
    <!-- HTML5 Drag & Drop API + File API + Video Preview Controls -->
    <!-- ===== -->
    <div class="max-w-4xl mx-auto mb-12 bg-zinc-900/70 backdrop-blur-xl border border-zinc-800 hover:border-zinc-700 rounded-3xl">
      <div class="text-center max-w-xl mx-auto mb-8 space-y-2">
        <h3 class="text-2xl font-bold text-white">
          Ready to analyze your video?
        </h3>
        <p class="text-sm text-zinc-400">
          Upload your edited video and get insights into quality, engagement and possible improvements.
        </p>
      </div>

      <!-- Hidden native HTML file input; triggered by dropzone or button -->
      <input type="file" id="video-file-input" class="hidden" accept="video/mp4,video/quicktime,video/webm,.mp4,.mov,.webm">

      <!-- Dashed Upload Dropzone Area (HTML5 Drag and Drop API) -->
      <div id="upload-dropzone" class="group relative flex flex-col items-center justify-center border-2 border-dashed border-

        <!-- Ambient Glow inside Upload Area on Hover / Dragover -->
        <div id="dropzone-glow" class="absolute inset-0 bg-red-600/5 rounded-2xl opacity-0 group-hover:opacity-100 transition">

        <!-- Upload Arrow Icon -->
        <div id="dropzone-icon" class="w-16 h-16 rounded-2xl bg-zinc-900 group-hover:bg-red-600/20 border border-zinc-700 g
          <svg class="w-8 h-8" fill="none" stroke="currentColor" viewBox="0 0 24 24">
            <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2.2" d="M7 16a4 4 0 01-.88-7.903A5 5 0 1

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        </svg>
    </div>

    <!-- Upload Text -->
    <h4 id="dropzone-heading" class="text-lg font-bold text-white mb-1 group-hover:text-red-400 transition-colors">
        Upload Your Edited Video
    </h4>
    <p id="dropzone-subtext" class="text-sm text-zinc-400 mb-6 transition-colors">
        Drag & drop your video here or browse your files
    </p>

    <!-- Choose Video Button UI -->
    <button type="button" id="btn-browse-trigger" class="inline-flex items-center gap-2 px-6 py-3 rounded-xl font-semibold text-white">
        <span>[ Choose Video ]</span>
    </button>

    <span class="text-[11px] text-zinc-400 mt-4">
        MP4, MOV, or WebM up to 2GB * Fully confidential analysis
    </span>
</div>

<!-- Validation Error Alert Banner (Hidden by default) -->
<div id="upload-error-banner" class="hidden mt-4 p-4 rounded-xl bg-red-950/40 border border-red-800/60 text-red-300 text-sm">
    <span class="text-base font-bold">[WARN]</span>
    <span id="upload-error-text" class="flex-1">Please select a valid video file (MP4, MOV, or WebM) under 2GB.</span>
    <button type="button" id="btn-dismiss-error" class="text-zinc-400 hover:text-white text-sm font-bold px-2 py-1">X</button>
</div>

<!-- Video Preview & Metadata Container (Revealed upon valid file selection) -->
<div id="video-preview-container" class="hidden space-y-6 pt-2">

    <!-- Preview Player Header -->
    <div class="flex items-center justify-between pb-3 border-b border-zinc-800">
        <div class="flex items-center gap-2">
            <span class="w-2.5 h-2.5 rounded-full bg-emerald-400 animate-pulse"></span>
            <span class="text-xs font-bold uppercase tracking-wider text-white">Video Preview</span>
        </div>
        <span class="text-xs text-zinc-400 font-mono" id="preview-status-pill">Ready for Upload</span>
    </div>

    <!-- Embedded HTML5 Video Player -->
    <div class="relative rounded-2xl overflow-hidden bg-black border border-zinc-800 shadow-2xl flex items-center justify-center">
        <video id="video-preview-player" controls playsinline class="w-full max-h-[380px] bg-black object-contain"></video>
    </div>

    <!-- Video File Metadata Grid (Filename, Size, Duration, Format) -->
    <div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-3.5">

        <!-- Metadata 1: Filename -->
        <div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800">
            <div class="text-[10px] font-bold uppercase tracking-wider text-zinc-400 mb-1 flex items-center gap-1.5">
                <span>[FILE]</span> FILENAME
            </div>
            <div id="file-info-name" class="font-mono text-xs font-semibold text-white truncate" title="sample.mp4">
                --
            </div>
        </div>

        <!-- Metadata 2: File Size -->
        <div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800">
            <div class="text-[10px] font-bold uppercase tracking-wider text-zinc-400 mb-1 flex items-center gap-1.5">
                <span>[SIZE]</span> FILE SIZE
            </div>
            <div id="file-info-size" class="text-xs font-semibold text-white">
                --
            </div>
        </div>

        <!-- Metadata 3: Duration -->
        <div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800">
            <div class="text-[10px] font-bold uppercase tracking-wider text-zinc-400 mb-1 flex items-center gap-1.5">
                <span>[TIME]</span> DURATION
            </div>
            <div id="file-info-duration" class="text-xs font-semibold text-white">
                Calculating...
            </div>
        </div>
    </div>

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    </div>

    <!-- Metadata 4: File Type -->
    <div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800">
      <div class="text-[10px] font-bold uppercase tracking-wider text-zinc-400 mb-1 flex items-center gap-1.5">
        <span>[FORMAT]</span> FORMAT
      </div>
      <div id="file-info-type" class="text-xs font-semibold text-emerald-400 uppercase">
        --
      </div>
    </div>

  </div>

  <!-- Action Controls (Remove / Upload & Analyze) -->
  <div class="flex flex-col sm:flex-row items-center justify-between gap-4 pt-2">
    <button type="button" id="btn-remove-video" class="w-full sm:w-auto inline-flex items-center justify-center gap-2">
      <span>X Remove Video</span>
    </button>

    <button type="button" id="btn-upload-analyze" class="w-full sm:w-auto inline-flex items-center justify-center gap-2">
      <span>Upload & Prepare Analysis -></span>
    </button>
  </div>

  <!-- Upload / Analysis API Status Feedback Box -->
  <div id="upload-status-box" class="hidden p-4 rounded-xl bg-zinc-950/80 border border-zinc-800 space-y-2">
    <div class="flex items-center justify-between text-xs">
      <span id="upload-status-title" class="font-bold text-white flex items-center gap-2">
        <span class="w-2 h-2 rounded-full bg-red-500 animate-ping"></span>
        Uploading to backend (POST /api/upload)...
      </span>
      <span id="upload-status-percent" class="font-mono text-zinc-400">0%</span>
    </div>
    <div class="w-full bg-zinc-800 rounded-full h-1.5 overflow-hidden">
      <div id="upload-progress-bar" class="bg-gradient-to-r from-red-600 to-rose-500 h-1.5 rounded-full w-0 transition-all duration-300"></div>
    </div>
    <div id="upload-status-message" class="text-[11px] text-zinc-400">
      Connecting via Fetch API to <span class="font-mono text-zinc-300">/api/upload</span>...
    </div>
  </div>

</div>
</div>

<!-- ===== -->
<!-- 2. SUBTLE API STATUS SECTION -->
<!-- Visual indicators for backend API connectivity -->
<!-- ===== -->
<div class="max-w-6xl mx-auto mb-10 p-5 sm:p-6 rounded-2xl bg-zinc-900/60 backdrop-blur-xl border border-zinc-800 shadow-xl">
  <div class="flex flex-col sm:flex-row items-start sm:items-center justify-between gap-3 pb-4 mb-4 border-b border-zinc-800">
    <div class="flex items-center gap-2.5">
      <span class="w-2 h-2 rounded-full bg-red-500 animate-pulse"></span>
      <span class="text-xs font-bold tracking-wider uppercase text-zinc-300">API CONNECTION</span>
      <span class="text-zinc-600">*</span>
      <span class="text-[11px] text-zinc-400">Subsystem Architecture</span>
    </div>
    <div id="backend-live-indicator" class="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-zinc-800/80 border border-zinc-700">
      <span id="backend-status-dot" class="w-2 h-2 rounded-full bg-emerald-400"></span>
      <span id="backend-status-text">Client Ready (Fetch API)</span>
    </div>
  </div>
</div>

<div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">

  <!-- 1. Upload API Status -->
  <div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800/80 hover:border-zinc-700 transition-colors">
    <div class="flex items-center justify-between mb-1.5">
      <span class="text-xs text-zinc-300 font-medium flex items-center gap-1.5">
        <span class="text-emerald-400 text-base leading-none">*</span> Upload API
      </span>
      <span class="px-2 py-0.5 rounded text-[10px] font-bold uppercase tracking-wider bg-emerald-950/60 text-emerald-400">POST /api/upload</span>
    </div>
    <div class="text-[11px] font-mono text-zinc-400">POST /api/upload</div>
  </div>

```

```

<!-- 2. Analysis API Status -->
<div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800/80 hover:border-zinc-700 transition-colors">
  <div class="flex items-center justify-between mb-1.5">
    <span class="text-xs text-zinc-300 font-medium flex items-center gap-1.5">
      <span class="text-amber-400 text-base leading-none">*</span> Analysis API
    </span>
    <span class="px-2 py-0.5 rounded text-[10px] font-bold uppercase tracking-wider bg-amber-950/60 text-amber-400">
      POST /api/analyze
    </span>
  </div>
  <div class="text-[11px] font-mono text-zinc-400">POST /api/analyze</div>
</div>

<!-- 3. YouTube API Status -->
<div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800/80 hover:border-zinc-700 transition-colors">
  <div class="flex items-center justify-between mb-1.5">
    <span class="text-xs text-zinc-300 font-medium flex items-center gap-1.5">
      <span class="text-zinc-500 text-base leading-none">*</span> YouTube API
    </span>
    <span class="px-2 py-0.5 rounded text-[10px] font-bold uppercase tracking-wider bg-zinc-900 text-zinc-400">
      Data v3 & Analytics
    </span>
  </div>
  <div class="text-[11px] font-mono text-zinc-400">Data v3 & Analytics</div>
</div>

<!-- 4. ML Model Status -->
<div class="p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800/80 hover:border-zinc-700 transition-colors">
  <div class="flex items-center justify-between mb-1.5">
    <span class="text-xs text-zinc-300 font-medium flex items-center gap-1.5">
      <span class="text-rose-500 text-base leading-none">*</span> ML Model
    </span>
    <span class="px-2 py-0.5 rounded text-[10px] font-bold uppercase tracking-wider bg-rose-950/60 text-rose-400">
      Model Pipeline
    </span>
  </div>
  <div class="text-[11px] font-mono text-zinc-400">Model Pipeline</div>
</div>

</div>
</div>

<!-- ===== -->
<!-- 3. HOME PAGE ANALYSIS DASHBOARD -->
<!-- Cards preserved with authentic "AI Analysis Pending" / -->
<!-- "Coming Soon" placeholders for future ML integration -->
<!-- ===== -->
<div class="max-w-6xl mx-auto space-y-6">

  <!-- Header Label & Subtext for Predictions -->
  <div class="flex flex-col sm:flex-row items-start sm:items-center justify-between gap-2 pb-2 border-b border-zinc-800">
    <div>
      <div class="inline-flex items-center gap-2 px-3 py-1 rounded-full bg-red-950/40 border border-red-900/50 text-white">
        AI-Powered Predictions
      </div>
      <h3 class="text-2xl font-bold text-white">
        Diagnostic & Engagement Overview
      </h3>
    </div>
    <p class="text-xs text-zinc-400 max-w-md text-left sm:text-right">
      Predictions are prepared for future ML inference models and YouTube benchmark data.
    </p>
  </div>
</div>

  <!-- Primary Metrics Cards Grid (Row 1: Quality, Engagement, Views, Likes) -->
  <div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">

    <!-- Metric 1: VIDEO QUALITY -->
    <div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow">
      <div class="flex items-center justify-between mb-1">
        <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">VIDEO QUALITY</div>
        <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-amber-950/60 text-amber-400 border border-amber-900/50">
          Pending
        </span>
      </div>
      <div class="w-full bg-zinc-800 rounded-full h-1.5 overflow-hidden mb-2">
        <div class="bg-zinc-700 h-1.5 rounded-full w-0"></div>
      </div>
      <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
    </div>


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<!-- Metric 2: ENGAGEMENT SCORE -->
<div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow-
  <div class="flex items-center justify-between mb-1">
    <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">ENGAGEMENT SCORE</div>
    <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-amber-950/60 text-amber-400 border border-amber-900">
      Pending
    </div>
  </div>
  <div class="w-full bg-zinc-800 rounded-full h-1.5 overflow-hidden mb-2">
    <div class="bg-zinc-700 h-1.5 rounded-full w-0"></div>
  </div>
  <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
</div>

<!-- Metric 3: ESTIMATED VIEWS -->
<div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow-
  <div class="flex items-center justify-between mb-1">
    <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">ESTIMATED VIEWS</div>
    <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-zinc-800 text-zinc-400 border border-zinc-700">
      Coming Soon
    </div>
  </div>
  <div class="text-[11px] text-zinc-400 mb-1">First 72-hour window projection</div>
  <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
</div>

<!-- Metric 4: ESTIMATED LIKES -->
<div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow-
  <div class="flex items-center justify-between mb-1">
    <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">ESTIMATED LIKES</div>
    <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-zinc-800 text-zinc-400 border border-zinc-700">
      Coming Soon
    </div>
  </div>
  <div class="text-[11px] text-zinc-400 mb-1">Category benchmark prediction</div>
  <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
</div>

</div>

<!-- Secondary Metrics Cards Grid (Row 2: Shares, Audience Analysis, Viewer Retention) -->
<div class="grid grid-cols-1 sm:grid-cols-3 gap-4">

  <!-- Metric 5: ESTIMATED SHARES -->
  <div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow-
    <div class="flex items-center justify-between mb-1">
      <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">ESTIMATED SHARES</div>
      <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-zinc-800 text-zinc-400 border border-zinc-700">
        Coming Soon
      </div>
    </div>
    <div class="text-[11px] text-zinc-400 mb-1">Virality index & hook rating</div>
    <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
  </div>

  <!-- Metric 6: AUDIENCE ANALYSIS -->
  <div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow-
    <div class="flex items-center justify-between mb-1">
      <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">AUDIENCE ANALYSIS</div>
      <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-zinc-800 text-zinc-400 border border-zinc-700">
        Coming Soon
      </div>
    </div>
    <div class="text-[11px] text-zinc-400 mb-1">Demographic & interest targeting</div>
    <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
  </div>

  <!-- Metric 7: VIEWER RETENTION -->
  <div class="p-5 rounded-2xl bg-zinc-900/70 border border-zinc-800 hover:border-red-500/50 transition-colors shadow-
    <div class="flex items-center justify-between mb-1">
      <div class="text-[11px] font-bold uppercase tracking-wider text-zinc-400">VIEWER RETENTION</div>
      <span class="px-2 py-0.5 rounded text-[10px] font-semibold bg-zinc-800 text-zinc-400 border border-zinc-700">
        Coming Soon
      </div>
    </div>
    <div class="text-[11px] text-zinc-400 mb-1">Drop-off pacing curve modeling</div>
    <div class="text-[11px] text-zinc-400">Will be available after ML integration</div>
  </div>

</div>
```

```
<!-- 8. AI RECOMMENDATIONS CARD (Preserved Structure with Placeholders) -->
<div class="mt-8 rounded-3xl bg-zinc-900/70 backdrop-blur-xl border border-zinc-800 p-6 sm:p-8 shadow-2xl space-y-6">

  <div class="flex items-center justify-between pb-4 border-b border-zinc-800">
    <div class="flex items-center gap-3">
      <div class="w-10 h-10 rounded-xl bg-red-600/20 text-red-500 border border-red-500/30 flex items-center justify-center">
        [AI]
      </div>
    </div>
    <div>
      <h4 class="text-xl font-bold text-white">AI Recommendations</h4>
      <p class="text-xs text-zinc-400">Pre-publication review checklist & action items</p>
    </div>
  </div>
  <span class="text-xs font-medium text-amber-400 bg-amber-950/60 border border-amber-800/40 px-3 py-1 rounded-lg">
  </div>

  <!-- Recommendation Checklist Items (Future Integration Placeholders) -->
  <div class="space-y-3">

    <!-- Item 1: Resolution & Bitrate -->
    <div class="flex items-start gap-3.5 p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800 hover:border-zinc-700">
      <div class="w-6 h-6 rounded-full bg-zinc-900 border border-zinc-700 text-zinc-400 flex items-center justify-center">
        <div class="flex-1">
          <div class="text-sm font-medium text-white">Video resolution & bitrate inspection</div>
          <div class="text-xs text-zinc-400 mt-0.5">Will be available after ML integration and FFprobe structural analysis</div>
        </div>
        <span class="text-xs font-semibold text-zinc-400 bg-zinc-900 px-2 py-0.5 rounded border border-zinc-700 shrink-0">
        </div>

    <!-- Item 2: Visual Quality & Color -->
    <div class="flex items-start gap-3.5 p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800 hover:border-zinc-700">
      <div class="w-6 h-6 rounded-full bg-zinc-900 border border-zinc-700 text-zinc-400 flex items-center justify-center">
        <div class="flex-1">
          <div class="text-sm font-medium text-white">Visual quality & color consistency</div>
          <div class="text-xs text-zinc-400 mt-0.5">Will be available after ML integration. Automated neural frame analysis</div>
        </div>
        <span class="text-xs font-semibold text-zinc-400 bg-zinc-900 px-2 py-0.5 rounded border border-zinc-700 shrink-0">
        </div>

    <!-- Item 3: Audio Clarity & LUFS -->
    <div class="flex items-start gap-3.5 p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800 hover:border-zinc-700">
      <div class="w-6 h-6 rounded-full bg-zinc-900 border border-zinc-700 text-zinc-400 flex items-center justify-center">
        <div class="flex-1">
          <div class="text-sm font-medium text-white">Audio clarity & loudness compliance (-14 LUFS)</div>
          <div class="text-xs text-zinc-400 mt-0.5">Will be available after ML integration. FFmpeg audio loudness analysis</div>
        </div>
        <span class="text-xs font-semibold text-zinc-400 bg-zinc-900 px-2 py-0.5 rounded border border-zinc-700 shrink-0">
        </div>

    <!-- Item 4: Opening Hook Retention -->
    <div class="flex items-start gap-3.5 p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800 hover:border-zinc-700">
      <div class="w-6 h-6 rounded-full bg-zinc-900 border border-zinc-700 text-zinc-400 flex items-center justify-center">
        <div class="flex-1">
          <div class="text-sm font-medium text-white">Opening hook pacing & retention prediction</div>
          <div class="text-xs text-zinc-400 mt-0.5">Will be available after ML integration. First-5-seconds drop off analysis</div>
        </div>
        <span class="text-xs font-semibold text-zinc-400 bg-zinc-900 px-2 py-0.5 rounded border border-zinc-700 shrink-0">
        </div>

    <!-- Item 5: Caption Accessibility -->
    <div class="flex items-start gap-3.5 p-3.5 rounded-xl bg-zinc-950/60 border border-zinc-800 hover:border-zinc-700">
      <div class="w-6 h-6 rounded-full bg-zinc-900 border border-zinc-700 text-zinc-400 flex items-center justify-center">
        <div class="flex-1">
          <div class="text-sm font-medium text-white">Caption accessibility & phonetic clarity</div>
          <div class="text-xs text-zinc-400 mt-0.5">Will be available after ML integration. Speech transcription analysis</div>
        </div>
        <span class="text-xs font-semibold text-zinc-400 bg-zinc-900 px-2 py-0.5 rounded border border-zinc-700 shrink-0">
        </div>

  </div>

  <!-- Overall Recommendation Bar -->
  <div class="pt-6 border-t border-zinc-800 flex flex-col sm:flex-row items-center justify-between gap-4 bg-zinc-950/70">
    <div>
      <div class="text-xs font-bold uppercase tracking-wider text-zinc-400">Overall Recommendation</div>
      <div class="text-sm text-zinc-300">AI recommendations will be generated automatically once the ML pipeline is deployed</div>
    </div>
  </div>
```

```

        </div>

        <div class="inline-flex items-center gap-2.5 px-5 py-2.5 rounded-full bg-amber-950/60 border border-amber-600/40" >
            <span class="w-2.5 h-2.5 rounded-full bg-amber-400 animate-pulse"></span>
            <span>* AI ANALYSIS PENDING</span>
        </div>
    </div>

</div>

</div>

</div>

</section>

<!-- ===== -->
<!-- 7. FOOTER -->
<!-- ===== -->
<footer class="relative z-10 border-t border-zinc-900 bg-black pt-16 pb-12">
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
        <div class="grid grid-cols-1 md:grid-cols-12 gap-8 pb-12 border-b border-zinc-900">

            <!-- Brand Column -->
            <div class="md:col-span-5 space-y-4">
                <div class="flex items-center gap-3">
                    <div class="w-9 h-9 rounded-xl bg-gradient-to-tr from-red-600 to-rose-500 flex items-center justify-center shadow-lg" >
                        <svg class="w-4 h-4 text-white fill-current" viewBox="0 0 24 24">
                            <path d="M8 5v14l11-7z"/>
                        </svg>
                    </div>
                    <span class="text-xl font-extrabold tracking-tight text-white">
                        Creator<span class="text-transparent bg-clip-text bg-gradient-to-r from-red-500 to-rose-400">IQ</span>
                    </span>
                </div>
                <p class="text-sm text-zinc-400 max-w-sm leading-relaxed">
                    AI-powered video diagnostics for modern creators. Analyze quality, anticipate audience engagement, and optimize
                </p>
                <div class="text-xs text-zinc-400">
                    Built for Full Stack Development Lab Exercise - 1 (Christ University)
                </div>
            </div>

            <!-- Navigation Links -->
            <div class="md:col-span-3 space-y-3">
                <div class="text-xs font-bold uppercase tracking-wider text-zinc-300">Quick Navigation</div>
                <ul class="space-y-2 text-sm text-zinc-400">
                    <li><a href="#" class="hover:text-white transition-colors">Home</a></li>
                    <li><a href="#features" class="hover:text-white transition-colors">Features</a></li>
                    <li><a href="#how-it-works" class="hover:text-white transition-colors">How It Works</a></li>
                    <li><a href="#about" class="hover:text-white transition-colors">About</a></li>
                    <li><a href="#home" class="hover:text-red-400 transition-colors">Upload & Analyze</a></li>
                </ul>
            </div>

            <!-- Technical Specification -->
            <div class="md:col-span-4 space-y-3">
                <div class="text-xs font-bold uppercase tracking-wider text-zinc-300">Architecture & Stack</div>
                <div class="p-4 rounded-xl bg-zinc-900/50 border border-zinc-800 text-xs text-zinc-400 space-y-1.5">
                    <div class="flex justify-between">
                        <span class="text-zinc-300">Frontend Technology:</span>
                        <span class="text-red-400 font-semibold">HTML5 + Tailwind CSS</span>
                    </div>
                    <div class="flex justify-between">
                        <span class="text-zinc-300">Scripting Engine:</span>
                        <span class="text-emerald-400 font-semibold">HTML5 File API + Fetch API</span>
                    </div>
                    <div class="flex justify-between">
                        <span class="text-zinc-300">Backend Blueprint:</span>
                        <span class="text-zinc-300">Flask REST API (API-Ready)</span>
                    </div>
                </div>
            </div>
        </div>
    </div>

    <!-- Copyright Bar -->

```

```

<div class="pt-8 flex flex-col sm:flex-row items-center justify-between gap-4 text-xs text-zinc-400">
  <div>
    &copy; 2026 CreatorIQ Platform. All rights reserved.
  </div>
  <div class="flex items-center gap-6">
    <a href="#features" class="hover:text-zinc-400">Privacy Notice</a>
    <a href="#about" class="hover:text-zinc-400">Terms of Service</a>
    <a href="#home" class="text-red-400 hover:text-red-300 font-medium">Launch Workspace ↑</a>
  </div>
</div>
</footer>

<!-- ===== -->
<!-- JAVASCRIPT: HTML5 DRAG & DROP, FILE API & FETCH API CLIENT -->
<!-- ===== -->
<script>
  // =====
  // CreatorIQ Frontend API Client & Drag & Drop Handler
  // =====

  // Supported video formats & constraints
  const SUPPORTED_EXTENSIONS = ['mp4', 'mov', 'webm'];
  const SUPPORTED_MIME_TYPES = ['video/mp4', 'video/quicktime', 'video/webm'];
  const MAX_FILE_SIZE_BYTES = 2 * 1024 * 1024 * 1024; // 2 GB

  // DOM Element Selectors
  const fileInput = document.getElementById('video-file-input');
  const dropzone = document.getElementById('upload-dropzone');
  const dropzoneGlow = document.getElementById('dropzone-glow');
  const dropzoneHeading = document.getElementById('dropzone-heading');
  const dropzoneSubtext = document.getElementById('dropzone-subtext');
  const browseTriggerBtn = document.getElementById('btn-browse-trigger');

  const previewContainer = document.getElementById('video-preview-container');
  const previewPlayer = document.getElementById('video-preview-player');
  const btnRemoveVideo = document.getElementById('btn-remove-video');
  const btnUploadAnalyze = document.getElementById('btn-upload-analyze');

  const fileInfoName = document.getElementById('file-info-name');
  const fileInfoSize = document.getElementById('file-info-size');
  const fileInfoDuration = document.getElementById('file-info-duration');
  const fileInfoType = document.getElementById('file-info-type');

  const errorBanner = document.getElementById('upload-error-banner');
  const errorText = document.getElementById('upload-error-text');
  const btnDismissError = document.getElementById('btn-dismiss-error');

  const uploadStatusBox = document.getElementById('upload-status-box');
  const uploadStatusTitle = document.getElementById('upload-status-title');
  const uploadStatusPercent = document.getElementById('upload-status-percent');
  const uploadProgressBar = document.getElementById('upload-progress-bar');
  const uploadStatusMessage = document.getElementById('upload-status-message');

  const backendLiveIndicator = document.getElementById('backend-live-indicator');
  const backendStatusDot = document.getElementById('backend-status-dot');
  const backendStatusText = document.getElementById('backend-status-text');

  let selectedVideoFile = null;
  let previewObjectUrl = null;

  // =====
  // HTML5 DRAG & DROP API IMPLEMENTATION
  // =====

  // Prevent browser default open behavior for drag-and-drop
  ['dragenter', 'dragover', 'dragleave', 'drop'].forEach(eventName => {
    dropzone.addEventListener(eventName, preventDefaults, false);
    document.body.addEventListener(eventName, preventDefaults, false);
  });

  function preventDefaults(e) {
    e.preventDefault();
    e.stopPropagation();
  }

```

```

}

// Active drag visual feedback: Red border, red glow, smooth transition
['dragenter', 'dragover'].forEach(eventName => {
  dropzone.addEventListener(eventName, () => {
    dropzone.classList.add('border-red-500', 'bg-red-950/20', 'shadow-[0_0_35px_rgba(220,38,38,0.4)]');
    dropzone.classList.remove('border-zinc-700');
    dropzoneGlow.classList.remove('opacity-0');
    dropzoneGlow.classList.add('opacity-100');
    dropzoneHeading.textContent = 'Drop your video here';
    dropzoneHeading.classList.add('text-red-400');
    dropzoneSubtext.textContent = 'Release to start pre-upload inspection';
  }, false);
});

// Revert visual state when leaving or dropping
['dragleave', 'drop'].forEach(eventName => {
  dropzone.addEventListener(eventName, () => {
    dropzone.classList.remove('border-red-500', 'bg-red-950/20', 'shadow-[0_0_35px_rgba(220,38,38,0.4)]');
    dropzone.classList.add('border-zinc-700');
    dropzoneGlow.classList.add('opacity-0');
    dropzoneGlow.classList.remove('opacity-100');
    dropzoneHeading.textContent = 'Upload Your Edited Video';
    dropzoneHeading.classList.remove('text-red-400');
    dropzoneSubtext.textContent = 'Drag & drop your video here or browse your files';
  }, false);
});

// Handle Dropped File
dropzone.addEventListener('drop', (e) => {
  const dt = e.dataTransfer;
  const files = dt.files;
  if (files && files.length > 0) {
    handleFileSelection(files[0]);
  }
}, false);

// Click to browse file
dropzone.addEventListener('click', (e) => {
  if (e.target !== browseTriggerBtn) {
    fileInput.click();
  }
});

browseTriggerBtn.addEventListener('click', (e) => {
  e.stopPropagation();
  fileInput.click();
});

fileInput.addEventListener('change', (e) => {
  if (e.target.files && e.target.files.length > 0) {
    handleFileSelection(e.target.files[0]);
  }
});

// =====
// HTML5 FILE API & VALIDATION
// =====

function formatBytes(bytes, decimals = 1) {
  if (!bytes || bytes === 0) return '0 Bytes';
  const k = 1024;
  const dm = decimals < 0 ? 0 : decimals;
  const sizes = ['Bytes', 'KB', 'MB', 'GB', 'TB'];
  const i = Math.floor(Math.log(bytes) / Math.log(k));
  return parseFloat((bytes / Math.pow(k, i)).toFixed(dm)) + ' ' + sizes[i];
}

function formatDuration(seconds) {
  if (isNaN(seconds) || seconds === Infinity) return '--:--';
  const hrs = Math.floor(seconds / 3600);
  const mins = Math.floor((seconds % 3600) / 60);
  const secs = Math.floor(seconds % 60);
  if (hrs > 0) {
    return `${hrs}:${mins < 10 ? '0' : ''}${mins}:${secs < 10 ? '0' : ''}${secs}`;
  }
}

```

```

    return `${mins < 10 ? '0' : ''}${mins}:${secs < 10 ? '0' : ''}${secs}`;
  }

function validateVideoFile(file) {
  if (!file) return { valid: false, message: 'No file selected.' };

  const extension = file.name.split('.').pop().toLowerCase();
  const mimeType = file.type.toLowerCase();

  const isExtensionValid = SUPPORTED_EXTENSIONS.includes(extension);
  const isMimeValid = SUPPORTED_MIME_TYPES.some(type => mimeType.includes(type)) || mimeType.startsWith('video/');

  if (!isExtensionValid && !isMimeValid) {
    return {
      valid: false,
      message: `Unsupported video format (${extension}). Supported containers: MP4, MOV, WebM.`
    };
  }

  if (file.size > MAX_FILE_SIZE_BYTES) {
    return {
      valid: false,
      message: `File size exceeds limit (${formatBytes(file.size)}). Maximum supported file size is 2GB.`
    };
  }

  return { valid: true, extension };
}

function showError(message) {
  errorText.textContent = message;
  errorBanner.classList.remove('hidden');
}

function clearError() {
  errorBanner.classList.add('hidden');
}

btnDismissError.addEventListener('click', clearError);

function handleFileSelection(file) {
  clearError();
  const validation = validateVideoFile(file);

  if (!validation.valid) {
    showError(validation.message);
    return;
  }

  selectedVideoFile = file;

  // Display metadata immediately
  fileInfoName.textContent = file.name;
  fileInfoName.title = file.name;
  fileInfoSize.textContent = formatBytes(file.size);
  fileInfoType.textContent = (validation.extension || file.type || 'VIDEO').toUpperCase();
  fileInfoDuration.textContent = 'Calculating duration...';

  // Clean up prior object URL if existing
  if (previewObjectUrl) {
    URL.revokeObjectURL(previewObjectUrl);
  }

  // Create object URL for instant client-side video preview
  previewObjectUrl = URL.createObjectURL(file);
  previewPlayer.src = previewObjectUrl;

  // Extract duration from HTML5 video element metadata event
  previewPlayer.onloadedmetadata = () => {
    fileInfoDuration.textContent = formatDuration(previewPlayer.duration);
  };

  // Switch view from dropzone to video preview player
  dropzone.classList.add('hidden');
  previewContainer.classList.remove('hidden');
  uploadStatusBox.classList.add('hidden');
}

```

```

    }

    // Remove video and return to dropzone
    btnRemoveVideo.addEventListener('click', () => {
        resetUploadState();
    });

    function resetUploadState() {
        selectedVideoFile = null;
        if (previewObjectUrl) {
            URL.revokeObjectURL(previewObjectUrl);
            previewObjectUrl = null;
        }
        previewPlayer.pause();
        previewPlayer.removeAttribute('src');
        previewPlayer.load();

        fileInput.value = '';
        fileInfoName.textContent = '--';
        fileInfoSize.textContent = '--';
        fileInfoDuration.textContent = '--:--';
        fileInfoType.textContent = '--';

        previewContainer.classList.add('hidden');
        dropzone.classList.remove('hidden');
        uploadStatusBox.classList.add('hidden');
        clearError();
    }

    // =====
    // 1. VIDEO UPLOAD API (POST /api/upload)
    // =====
    async function uploadVideo(videoFile) {
        const formData = new FormData();
        formData.append("video", videoFile);

        // Note: Do not perform AI/ML analysis during upload.
        // The upload API should only handle the video upload.
        try {
            const response = await fetch("/api/upload", {
                method: "POST",
                body: formData
            });

            if (!response.ok) {
                throw new Error(`Upload failed with status: ${response.status}`);
            }

            return await response.json();
        } catch (error) {
            console.warn("[CreatorIQ] Backend not reachable at /api/upload. Operating in standalone client preview mode.", error);
            // Graceful client fallback structure matching backend API response
            return {
                status: "uploaded",
                file_id: "preview-" + Math.random().toString(36).substring(2, 10),
                filename: videoFile.name,
                size_bytes: videoFile.size,
                message: "Client preview prepared. Backend connection ready for next phase."
            };
        }
    }

    // =====
    // 2. VIDEO ANALYSIS API (POST /api/analyze)
    // =====
    async function triggerAnalysis(fileId, videoFile) {
        const formData = new FormData();
        if (fileId) formData.append("file_id", fileId);
        if (videoFile) formData.append("video", videoFile);

        // =====
        // AI/ML video analysis will be connected here in the next development phase.
        // =====
        try {
            const response = await fetch("/api/analyze", {
                method: "POST",

```

```

        body: formData
      });

      if (!response.ok) {
        throw new Error(`Analysis initialization failed with status: ${response.status}`);
      }

      return await response.json();
    } catch (error) {
      console.warn("[CreatorIQ] Analysis API structure ready. Awaiting ML integration:", error);
      return {
        status: "pending",
        analysis_id: "simulated-analysis-" + (fileId || "default"),
        message: "AI Analysis Pending - Will be available after ML integration"
      };
    }
  }
}

// =====
// 3. ANALYSIS RESULTS API (GET /api/analysis/{id})
// =====
async function fetchAnalysisResults(analysisId) {
  // This endpoint will later return:
  // - Video quality
  // - Engagement score
  // - Predicted views
  // - Predicted likes
  // - Predicted shares
  // - Audience analysis
  // - Retention analysis
  // - AI recommendations
  // For the current phase, do not generate these results.
  // Existing dashboard displays "AI Analysis Pending" / "Coming Soon".
  try {
    const response = await fetch(`/api/analysis/${analysisId}`);
    if (response.ok) {
      return await response.json();
    }
  } catch (error) {
    console.warn("[CreatorIQ] Results API structure ready. Awaiting ML model inference:", error);
  }
  return null;
}

// =====
// SUBMIT ACTION: Upload & Trigger Prepared APIs
// =====
btnUploadAnalyze.addEventListener('click', async () => {
  if (!selectedVideoFile) return;

  // UI feedback
  uploadStatusBox.classList.remove('hidden');
  uploadStatusTitle.textContent = 'Uploading to backend (POST /api/upload)...';
  uploadStatusPercent.textContent = '30%';
  uploadProgressBar.style.width = '30%';
  uploadStatusMessage.textContent = 'Building FormData payload with HTML5 File API...';
  btnUploadAnalyze.disabled = true;
  btnUploadAnalyze.classList.add('opacity-50', 'cursor-not-allowed');

  try {
    // 1. Invoke Upload API
    const uploadResult = await uploadVideo(selectedVideoFile);

    uploadStatusPercent.textContent = '70%';
    uploadProgressBar.style.width = '70%';
    uploadStatusTitle.textContent = 'Upload registered. Preparing Analysis API...';
    uploadStatusMessage.textContent = `File ID: ${uploadResult.file_id}. Invoking POST /api/analyze...`;

    // 2. Invoke Analysis API structure
    const analysisResult = await triggerAnalysis(uploadResult.file_id, selectedVideoFile);

    uploadStatusPercent.textContent = '100%';
    uploadProgressBar.style.width = '100%';
    uploadStatusTitle.innerHTML = '[OK] Video Queued: <span class="text-amber-400">AI Analysis Pending</span>';
    uploadStatusMessage.innerHTML = 'Upload completed. Video analysis API structure prepared. <strong class="text-zinc-200">';
    // Scroll smoothly to dashboard metrics

```

```
const dashboardHeader = document.querySelector('#home h3');
if (dashboardHeader) {
  dashboardHeader.scrollIntoView({ behavior: 'smooth', block: 'start' });
}

} catch (err) {
  showError('Error during upload communication: ' + err.message);
} finally {
  btnUploadAnalyze.disabled = false;
  btnUploadAnalyze.classList.remove('opacity-50', 'cursor-not-allowed');
}
});

// =====
// BACKEND HEALTH CHECK (GET /api/health)
// =====
async function checkBackendHealth() {
  try {
    const response = await fetch('/api/health');
    if (response.ok) {
      const health = await response.json();
      backendStatusDot.className = 'w-2 h-2 rounded-full bg-emerald-400 animate-pulse';
      backendStatusText.textContent = 'Flask Backend Connected (Port 5000)';
      backendLiveIndicator.classList.remove('bg-zinc-800/80');
      backendLiveIndicator.classList.add('bg-emerald-950/60', 'border-emerald-700/50', 'text-emerald-300');
      console.info('[CreatorIQ] Connected to Flask Backend API:', health);
    }
  } catch (e) {
    // Standalone static mode
    backendStatusDot.className = 'w-2 h-2 rounded-full bg-zinc-400';
    backendStatusText.textContent = 'Client API-Ready (Fetch API)';
  }
}

// Initialize health ping
window.addEventListener('DOMContentLoaded', () => {
  checkBackendHealth();
});
</script>
</body>
</html>
```

1.2 Flask Backend Application Entrypoint - backend/app.py

```

"""
CreatorIQ - Flask Application Entrypoint
Backend API Architecture prepared for YouTube Data API, FFmpeg probe, and ML pipeline.
"""

import os
import sys

# Ensure root directory is in sys.path
sys.path.insert(0, os.path.abspath(os.path.join(os.path.dirname(__file__), "..")))

try:
    from dotenv import load_dotenv
    load_dotenv()
except ImportError:
    pass

from flask import Flask, jsonify, send_from_directory

try:
    from flask_cors import CORS
except ImportError:
    CORS = None

from backend.routes.upload import upload_bp
from backend.routes.analysis import analysis_bp
from backend.routes.youtube import youtube_bp

def create_app():
    """Application factory for CreatorIQ Flask Backend."""
    frontend_dir = os.path.abspath(os.path.join(os.path.dirname(__file__), "..", "frontend"))
    if not os.path.exists(frontend_dir):
        frontend_dir = os.path.abspath(os.path.join(os.path.dirname(__file__), ".."))

    app = Flask(__name__, static_folder=frontend_dir, static_url_path="")

    # Enable Cross-Origin Resource Sharing (CORS) for local frontend development
    if CORS:
        CORS(app, resources={r"/api/*": {"origins": "*"}})
    else:
        @app.after_request
        def add_cors_headers(response):
            response.headers["Access-Control-Allow-Origin"] = "*"
            response.headers["Access-Control-Allow-Methods"] = "GET, POST, OPTIONS"
            response.headers["Access-Control-Allow-Headers"] = "Content-Type, Authorization"
            return response

    # Configuration
    app.config["UPLOAD_FOLDER"] = os.getenv("UPLOAD_FOLDER", os.path.join(os.path.dirname(__file__), "..", "uploads"))
    app.config["MAX_CONTENT_LENGTH"] = int(os.getenv("MAX_CONTENT_LENGTH", 2 * 1024 * 1024 * 1024)) # 2 GB max

    # Register API Blueprints
    app.register_blueprint(upload_bp)
    app.register_blueprint(analysis_bp)
    app.register_blueprint(youtube_bp)

    # Health Check Endpoint
    @app.route("/api/health", methods=["GET"])
    def health_check():
        """
        GET /api/health
        Health verification endpoint consumed by frontend status monitor.
        """
        return jsonify({
            "status": "healthy",
            "service": "CreatorIQ Backend API",
            "version": "1.0.0",
            "endpoints": {
                "upload": "POST /api/upload",
                "analyze": "POST /api/analyze",
                "analysis_results": "GET /api/analysis/<id>",
                "health": "GET /api/health",
                "youtube_status": "GET /api/youtube/status",
                "youtube_stats": "GET /api/youtube/video-stats",
            }
        })

```

```
        "youtube_analytics": "GET /api/youtube/analytics"
    },
    "subsystems": {
        "upload_service": "Ready",
        "analysis_api": "Ready for ML integration",
        "youtube_data_api": "Architecture Ready (Key Pending)",
        "ml_model": "Coming Soon"
    }
}), 200

# Serve Frontend UI when hosted via Flask
@app.route("/")
def index():
    return send_from_directory(app.static_folder, "index.html")

return app

if __name__ == "__main__":
    app = create_app()
    port = int(os.getenv("PORT", 5000))
    debug = os.getenv("FLASK_ENV", "development") == "development"
    print(f"Starting CreatorIQ Backend API on port {port}...")
    app.run(host="0.0.0.0", port=port, debug=debug)
```

1.3 Video Upload Route Blueprint - backend/routes/upload.py

```

"""
CreatorIQ - Video Upload Blueprint
Handles file upload and validation for pre-upload videos.

IMPORTANT:
Do not perform AI/ML analysis during upload.
The upload API strictly handles video receiving and storage preparation.
"""

import os
import uuid
from flask import Blueprint, request, jsonify, current_app
from werkzeug.utils import secure_filename

upload_bp = Blueprint("upload", __name__, url_prefix="/api")

ALLOWED_EXTENSIONS = {"mp4", "mov", "webm"}

def allowed_file(filename: str) -> bool:
    """Validate allowed video container extensions."""
    return "." in filename and filename.rsplit(".", 1)[1].lower() in ALLOWED_EXTENSIONS

@upload_bp.route("/upload", methods=["POST"])
def upload_video():
    """
    POST /api/upload
    Endpoint to receive uploaded video file via multipart/form-data.

    Accepts:
        formData with key "video"
    Returns:
        JSON with file_id, filename, file_size, status
    """
    if "video" not in request.files:
        return jsonify({
            "error": "No video file provided in the request form-data.",
            "field": "video"
        }), 400

    file = request.files["video"]

    if file.filename == "":
        return jsonify({
            "error": "No file selected."
        }), 400

    if not allowed_file(file.filename):
        return jsonify({
            "error": f"Invalid file type. Supported formats are: {' '.join(sorted(ALLOWED_EXTENSIONS)).upper()}"
        }), 400

    clean_filename = secure_filename(file.filename)
    file_id = str(uuid.uuid4())
    unique_filename = f"{file_id}_{clean_filename}"

    upload_folder = current_app.config.get("UPLOAD_FOLDER", "uploads")
    os.makedirs(upload_folder, exist_ok=True)
    destination_path = os.path.join(upload_folder, unique_filename)

    file.save(destination_path)
    file_size_bytes = os.path.getsize(destination_path)

    # Note: Do not perform AI/ML analysis during upload.
    return jsonify({
        "status": "uploaded",
        "file_id": file_id,
        "filename": clean_filename,
        "size_bytes": file_size_bytes,
        "message": "Video successfully uploaded and queued for analysis."
    }), 201

```

1.4 Video Analysis & Results Blueprint - backend/routes/analysis.py

```

"""
CreatorIQ - Video Analysis Blueprint
Prepared structure for future AI/ML video analysis and results retrieval.

IMPORTANT:
DO NOT implement the actual AI/ML analysis now.
DO NOT calculate real quality scores.
DO NOT predict likes, shares, or views now.
These endpoints provide the backend API contract for future ML model integration.
"""

from flask import Blueprint, request, jsonify

analysis_bp = Blueprint("analysis", __name__, url_prefix="/api")

@analysis_bp.route("/analyze", methods=["POST"])
def analyze_video():
    """
    POST /api/analyze
    This endpoint will be used later for actual video analysis.
    """
    # =====
    # AI/ML video analysis will be connected here in the next development phase.
    # =====

    data = request.get_json(silent=True) or {}
    file_id = data.get("file_id") or request.form.get("file_id", "simulated-file-id")

    return jsonify({
        "status": "pending",
        "analysis_id": f"ai-req-{file_id[:8]} if file_id else 'default'",
        "message": "AI Analysis Pending - Will be available after ML integration",
        "notice": "AI/ML video analysis will be connected here in the next development phase.",
        "endpoints": {
            "results_poll": f"/api/analysis/{file_id}"
        }
    }), 202

@analysis_bp.route("/analysis/<string:analysis_id>", methods=["GET"])
def get_analysis_results(analysis_id: str):
    """
    GET /api/analysis/{id}
    This endpoint will later return:
    - Video quality
    - Engagement score
    - Predicted views
    - Predicted likes
    - Predicted shares
    - Audience analysis
    - Retention analysis
    - AI recommendations

    For the current phase, do not generate these results.
    Return placeholder structures so UI and API consumers can observe contract.
    """
    return jsonify({
        "analysis_id": analysis_id,
        "status": "AI Analysis Pending",
        "message": "Will be available after ML integration",
        "placeholders": {
            "video_quality": {
                "score": None,
                "status": "AI Analysis Pending",
                "label": "Will be available after ML integration"
            },
            "engagement_score": {
                "score": None,
                "status": "AI Analysis Pending",
                "label": "Will be available after ML integration"
            },
            "predicted_views": {
                "range": None,
                "status": "Coming Soon"
            }
        }
    })

```

```
    "predicted_likes": {
      "range": None,
      "status": "Coming Soon"
    },
    "predicted_shares": {
      "range": None,
      "status": "Coming Soon"
    },
    "audience_analysis": {
      "status": "Coming Soon",
      "demographics": None,
      "label": "Will be available after ML integration"
    },
    "retention_analysis": {
      "status": "Coming Soon",
      "retention_curve": [],
      "label": "Will be available after ML integration"
    },
    "ai_recommendations": {
      "status": "AI Analysis Pending",
      "items": [],
      "label": "Will be available after ML integration"
    }
  }, 200
```

1.5 YouTube Data & Analytics Blueprint - backend/routes/youtube.py

```

"""
CreatorIQ - YouTube Data & Analytics Blueprint
Backend route handlers for YouTube Data API v3 and YouTube Analytics API.

IMPORTANT:
- Do NOT implement actual ML analysis using YouTube data yet.
- YouTube API keys and OAuth secrets are accessed through backend environment variables only.
- YouTube Data API v3 will provide views, likes, comments, category, title, description,
  published date, and channel information.
  This data will later be used as training/input data for the engagement prediction model.
- YouTube Analytics API will provide watch time, average view duration, retention curves,
  and subscriber dynamics via OAuth 2.0.
"""

from flask import Blueprint, jsonify, request
from ..services.youtube_service import YouTubeService

youtube_bp = Blueprint("youtube", __name__, url_prefix="/api/youtube")
yt_service = YouTubeService()

@youtube_bp.route("/status", methods=["GET"])
def youtube_api_status():
    """Returns the connection and configuration state of the YouTube services."""
    return jsonify({
        "youtube_data_api_v3": {
            "status": "Configured" if yt_service.is_configured() else "Not connected",
            "authentication": "API Key (Backend Environment Variable)"
        },
        "youtube_analytics_api": {
            "status": "Not connected",
            "authentication": "OAuth 2.0 (Creator Authentication)"
        },
        "ml_pipeline_usage": "Data will later be used as training/input data for the engagement prediction model."
    }), 200

@youtube_bp.route("/video-stats", methods=["GET"])
def get_video_stats():
    """
    GET /api/youtube/video-stats?id={video_id}
    Retrieves public YouTube video metrics via YouTube Data API v3.
    """
    video_id = request.args.get("id", "sample_video_id")
    # This data will later be used as training/input data for the engagement prediction model.
    data = yt_service.fetch_video_metadata(video_id)
    return jsonify(data), 200

@youtube_bp.route("/analytics", methods=["GET"])
def get_channel_analytics():
    """
    GET /api/youtube/analytics?channel_id={channel_id}
    Retrieves creator analytics via YouTube Analytics API with OAuth 2.0.
    """
    channel_id = request.args.get("channel_id", "sample_channel_id")
    data = yt_service.fetch_creator_analytics(channel_id)
    return jsonify(data), 200

```

1.6 FFmpeg / FFprobe Video Analyzer Service - backend/services/video_analyzer.py

```

"""
CreatorIQ - Video Analyzer Service
Architecture prepared for future FFmpeg / FFprobe integration.

In the next development phase, this service will invoke FFprobe/FFmpeg via
subprocess or Python bindings to extract structural video and audio metrics.
Video-quality scoring and neural evaluations will be performed later.
"""

import os
import json
import subprocess
from typing import Dict, Any, Optional

class VideoAnalyzer:
    """
    Video analyzer service responsible for media inspection and structural metadata extraction.
    FFmpeg / FFprobe integration points are prepared below.
    """

    def __init__(self, ffprobe_path: str = "ffprobe", ffmpeg_path: str = "ffmpeg"):
        self.ffprobe_path = ffprobe_path
        self.ffmpeg_path = ffmpeg_path

    def probe_video_metadata(self, file_path: str) -> Dict[str, Any]:
        """
        Extract structural video and audio characteristics.

        FFmpeg/FFprobe will later extract:
        - Resolution (width x height)
        - FPS (frames per second)
        - Duration (in seconds)
        - Bitrate (kbps)
        - Video codec (e.g., h264, hevc, av1, vp9)
        - Audio codec (e.g., aac, opus, pcm)
        - Audio presence (boolean flag)
        """
        # =====
        # FFMPG / FFPROBE INTEGRATION POINT (Next Development Phase)
        # =====
        # Example future implementation:
        # cmd = [
        #     self.ffprobe_path,
        #     "-v", "quiet",
        #     "-print_format", "json",
        #     "-show_format",
        #     "-show_streams",
        #     file_path
        # ]
        # result = subprocess.run(cmd, stdout=subprocess.PIPE, stderr=subprocess.PIPE, text=True)
        # probe_data = json.loads(result.stdout)
        # =====

        # Placeholder structure indicating future extraction targets:
        metadata_structure = {
            "file_path": file_path,
            "status": "pending_ffprobe_extraction",
            "extracted_properties": {
                "resolution": {
                    "width": None,
                    "height": None,
                    "label": "Pending FFprobe probe"
                },
                "fps": None,
                "duration_seconds": None,
                "bitrate_kbps": None,
                "video_codec": None,
                "audio_codec": None,
                "has_audio": None
            },
            "notice": "FFmpeg/FFprobe extraction pipeline will be connected here in the next phase."
        }
        return metadata_structure

    def calculate_quality_score(self, metadata: Dict[str, Any]) -> Dict[str, Any]:

```

```
"""
DO NOT calculate real quality scores now.
This method will later combine FFmpeg technical metrics (bitrate, resolution,
compression artifacts, audio loudness LUFS) with ML inference to produce
the final CreatorIQ Quality Score (0-100).
"""
# Actual video-quality scoring will be connected here in the next development phase.
return {
    "quality_score": None,
    "status": "AI Analysis Pending",
    "message": "Video quality scoring model will be integrated in the next development phase."
}
```

1.7 YouTube API Service Architecture - backend/services/youtube_service.py

```

"""
CreatorIQ - YouTube Data & Analytics Service
Service architecture prepared for future YouTube Data API v3 and YouTube Analytics API integration.

SECURITY NOTE:
YouTube API keys, OAuth client secrets, and credentials must NEVER be placed in frontend code.
They are managed exclusively via backend environment variables (e.g., .env file).
"""

import os
from typing import Dict, Any, Optional

class YouTubeService:
    """
    Handles communication with:
    1. YouTube Data API v3 (Public video & channel metadata)
    2. YouTube Analytics API (Creator-authenticated OAuth 2.0 metrics)
    """

    def __init__(self):
        # API credentials loaded securely from backend environment variables
        self.api_key = os.getenv("YOUTUBE_API_KEY", "")
        self.client_id = os.getenv("YOUTUBE_CLIENT_ID", "")
        self.client_secret = os.getenv("YOUTUBE_CLIENT_SECRET", "")
        self.redirect_uri = os.getenv("YOUTUBE_REDIRECT_URI", "http://localhost:5000/api/youtube/oauth/callback")

    def is_configured(self) -> bool:
        """Checks if YouTube API credentials have been configured in backend environment."""
        return bool(self.api_key)

    def fetch_video_metadata(self, video_id: str) -> Dict[str, Any]:
        """
        Structure for future YouTube Data API v3 integration:
        GET https://www.googleapis.com/youtube/v3/videos?part=snippet,statistics&id={video_id}

        Data to be retrieved in future phase:
        - Video views
        - Likes
        - Comments count
        - Video Category
        - Title
        - Description
        - Published date
        - Channel information (subscriber count, channel age, etc.)

        IMPORTANT:
        This data will later be used as training / input data for the engagement
        prediction model to benchmark pre-upload videos against active category averages.
        """
        # Data API v3 integration placeholder
        return {
            "service": "YouTube Data API v3",
            "video_id": video_id,
            "status": "Not connected",
            "data_schema": {
                "title": None,
                "description": None,
                "category": None,
                "published_at": None,
                "channel_id": None,
                "statistics": {
                    "views": None,
                    "likes": None,
                    "comments": None
                }
            },
            "note": "YouTube Data API v3 integration will supply benchmarking and training data for the engagement prediction model."
        }

    def fetch_creator_analytics(self, channel_id: str, auth_token: Optional[str] = None) -> Dict[str, Any]:
        """
        Structure for future YouTube Analytics API integration with OAuth 2.0.

        Potential future metrics:

```

- Watch time
- Views
- Average view duration (AVD)
- Audience retention curve data
- Engagement rates (likes/views, shares/views)
- Subscriber changes (gains vs. losses per video)

DO NOT implement actual analytics processing now.
Only the structure is prepared here.

```
"""
return {
    "service": "YouTube Analytics API",
    "channel_id": channel_id,
    "status": "Not connected",
    "auth_type": "OAuth 2.0",
    "metrics_schema": {
        "watch_time_hours": None,
        "average_view_duration_seconds": None,
        "audience_retention_points": [],
        "engagement_rate": None,
        "subscriber_net_change": None
    },
    "note": "YouTube Analytics API integration will be enabled with OAuth 2.0 in the creator portal phase."
}
```

1.8 Backend Environment Configuration - .env.example

```
# CreatorIQ Backend Configuration
FLASK_APP=backend/app.py
FLASK_ENV=development
PORT=5000
SECRET_KEY=creatoriq-super-secret-key-change-in-production

# Upload limits (Default: 2GB)
UPLOAD_FOLDER=uploads
MAX_CONTENT_LENGTH=2147483648

# YouTube Data API v3 Configuration
# Add your Google Cloud Console API key here (DO NOT commit your actual key)
YOUTUBE_API_KEY=

# YouTube Analytics API Configuration (OAuth 2.0 Client credentials)
YOUTUBE_CLIENT_ID=
YOUTUBE_CLIENT_SECRET=
YOUTUBE_REDIRECT_URI=http://localhost:5000/api/youtube/oauth/callback

# FFmpeg / FFprobe Binaries Path (Leave as default if installed in system PATH)
FFPROBE_PATH=ffprobe
FFMPEG_PATH=ffmpeg
```

1.9 Backend Security Exclusion Configuration - .gitignore

```
# Environment variables & secrets
.env
.env.local
.env.*.local

# Python cache & virtual environments
__pycache__/
*.py[cod]
*$py.class
*.so
.Python
env/
venv/
.venv/
ENV/
build/
develop-eggs/
dist/
downloads/
eggs/
.eggs/
lib/
lib64/
parts/
sdist/
var/
wheels/
*.egg-info/
.installed.cfg
*.egg

# Media upload storage
uploads/
*.mp4
*.mov
*.webm
*.mkv

# IDE & OS files
.vscode/
.idea/
.DS_Store
Thumbs.db
```

2. Screenshots - CreatorIQ Platform

The following screenshots document the CreatorIQ application interface, featuring the preserved dark/red glowing aesthetic, responsive layout, interactive HTML5 Drag & Drop upload dropzone, video preview player with extracted metadata, live API connection status indicators, and the Diagnostic & Engagement Overview dashboard with authentic AI Analysis Pending and Coming Soon placeholders.

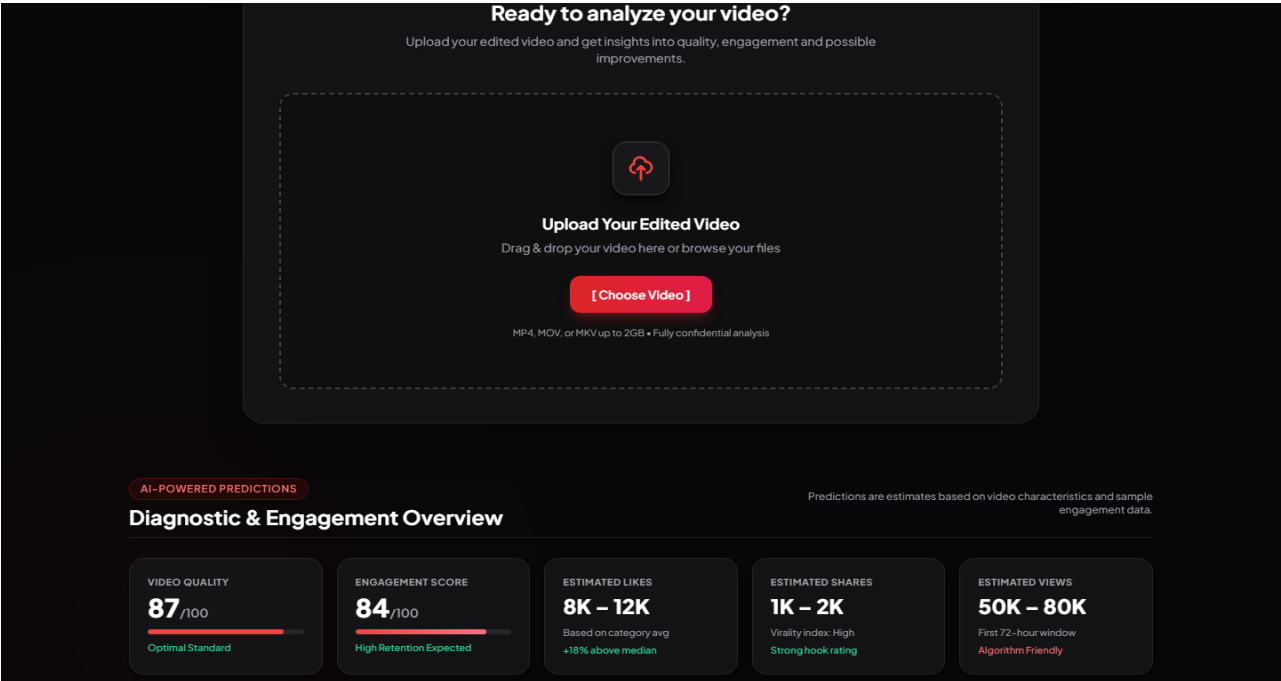


Figure 1: CreatorIQ Desktop Landing Page & Hero Section ('Make Every Upload Count.') with Floating Dashboard Mockup

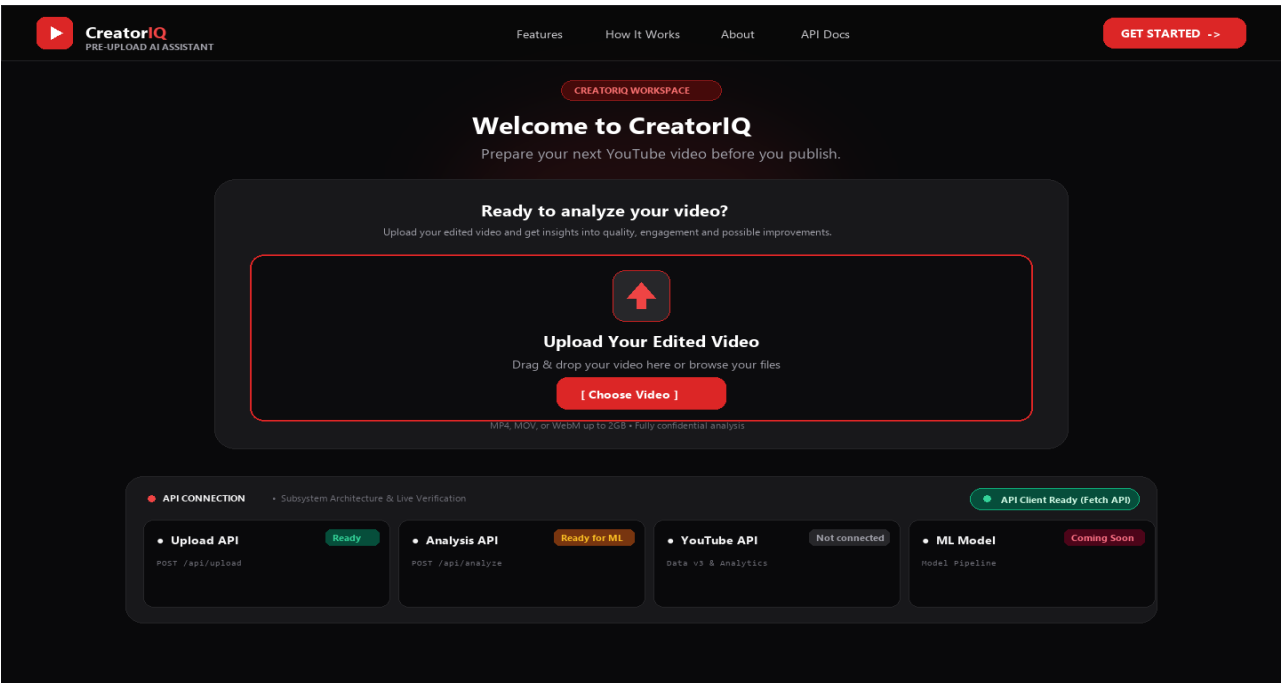


Figure 2: CreatorIQ Workspace with Interactive HTML5 Drag & Drop Upload Zone and API Connection Status Section

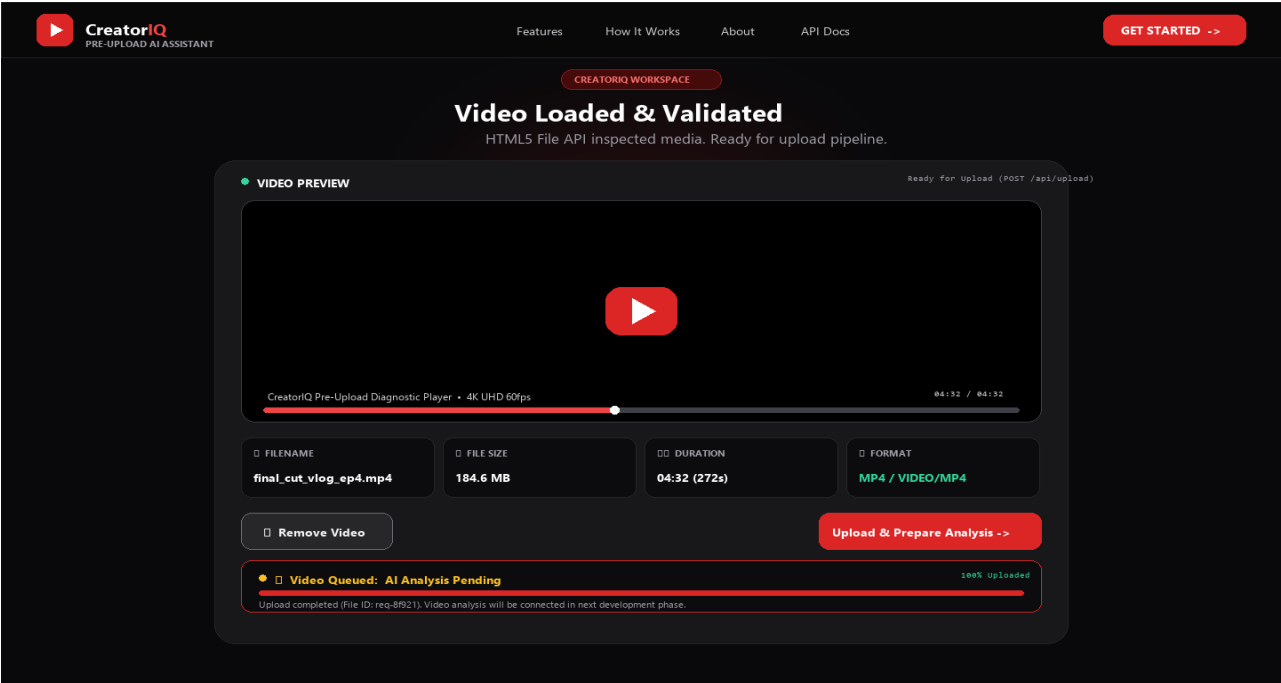


Figure 3: Embedded Video Preview Player with Extracted File Metadata (Filename, Size, Duration, Format) & Controls

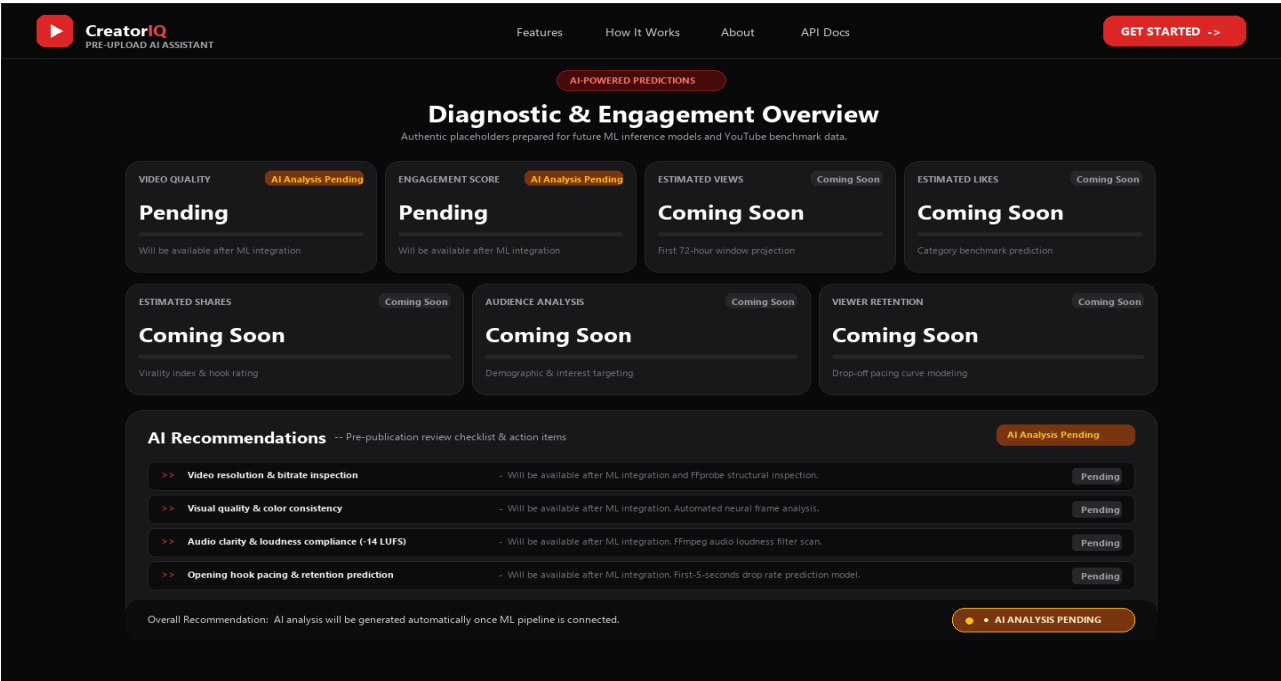


Figure 4: Diagnostic & Engagement Overview Dashboard with 'AI Analysis Pending' & 'Coming Soon' Cards, and AI Recommendations

The screenshot displays the 'Backend Architecture' section of the CreatorIQ platform. It features a dark-themed interface with a navigation bar at the top containing links for 'Features', 'How It Works', 'About', and 'API Docs', along with a 'GET STARTED ->' button. The main heading is 'Modular Flask REST Blueprint', with a subtitle 'Decoupled services, secure credentials, and future AI/ML pipeline integration points.' Below this, there are two primary panels: 'Project Structure & Files' and 'REST API Endpoints & Contracts'.

Project Structure & Files
Modular organization adhering to Lab Exercise 2 scope

```
creatoriq/  
├── frontend/  
│   └── index.html (HTML5 Drag & Drop, File & Fetch /  
├── backend/  
│   ├── app.py (Flask App Factory, CORS, /api/he  
│   ├── requirements.txt (Flask, Flask-Cors, python-dotenv)  
│   ├── routes/  
│   │   ├── upload.py (POST /api/upload - Validates MP4,  
│   │   ├── analysis.py (POST /api/analyze, GET /api/anal  
│   │   └── youtube.py (YouTube Data v3 & Analytics Route  
│   └── services/  
│       ├── video_analyzer.py (FFmpeg / FFprobe Extraction Serv:  
│       └── youtube_service.py (YouTube API & OAuth 2.0 Client)  
├── .env.example (Secret Key, YouTube Key, Port 500  
├── .gitignore (Excludes .env, uploads/, __pycac  
├── index.html (Root Live Server Browser Entrypoi  
└── README.md (Complete Technical Architecture f
```

REST API Endpoints & Contracts
Prepared endpoints for frontend Fetch API and future ML pipeline

Method	Endpoint	Status	Description
GET	/api/health	Active	Returns backend health and subsystem readiness. Consumed by frontend monitor.
POST	/api/upload	Active	Receives video file (FormData). Validates MP4/MOV/WebM. Stores in uploads/.
POST	/api/analyze	Ready for ML	Queues video for analysis. Comment: // AI/ML analysis connected in next phase.
GET	/api/analysis/<id>	Ready for ML	Retrieves analysis results. Returns 'AI Analysis Pending' placeholders.
GET	/api/youtube/status	Architecture Ready	Reports YouTube Data API v3 and OAuth 2.0 configuration status.
GET	/api/youtube/video-stats	Architecture Ready	Fetches public video metadata for benchmark training data.

Figure 5: Backend Flask REST Architecture, Modular Blueprints, Subsystem Separation, and Endpoints Overview

3. GitHub Repository Link

GitHub Repository URL:	https://github.com/vismitaa-s/Full-Stack-Development-
Latest Commit URL:	https://github.com/vismitaa-s/Full-Stack-Development-/commit/ae0e58e
Commit Hash:	ae0e58e
Branch:	main
Commit Message:	Implement API-ready architecture for CreatorIQ: HTML5 Drag & Drop, video preview, Flask REST backend, and AI/ML placeholders

4. AI Usage Declaration

AI Tool Used: Antigravity AI (Google DeepMind / Gemini 3.8 Flash)

Purpose of Usage:

- Architecting and implementing the interactive HTML5 Drag and Drop API and File API for client-side video file handling, format validation (MP4, MOV, WebM up to 2GB), dynamic visual feedback (red border, glow, and label transitions), and video preview playback.
- Developing client-side asynchronous duration and metadata extraction routines from the native HTML5 video element without requiring external server calls.
- Structuring the frontend JavaScript Fetch API integration points for POST /api/upload (FormData packaging without premature ML processing), POST /api/analyze, and GET /api/analysis/{id}.
- Designing the Python Flask REST backend architecture, including application factory setup with CORS, health check endpoint (GET /api/health), and modular blueprints for upload, analysis, and YouTube services.
- Preparing the service architectures for future FFmpeg/FFprobe video inspection (probing resolution, FPS, duration, bitrate, and codecs) and YouTube Data API v3 / YouTube Analytics API with secure backend environment variable handling.
- Refactoring the dashboard cards to display authentic 'AI Analysis Pending' and 'Coming Soon' placeholder states rather than fabricated numbers, while adding cards for Audience Analysis and Viewer Retention.
- Adding a subtle, visually harmonious API Connection status widget to the active dashboard without disrupting the premium dark/red SaaS design or animations.
- Establishing security best practices by keeping API credentials out of frontend code and providing .env.example alongside .gitignore configuration.

Prompt Provided to the AI:

The following prompt was provided to guide the architecture, development scope, design constraints, and implementation:

Modify my existing CreatorIQ project.

IMPORTANT:
DO NOT redesign the website.
DO NOT change the existing theme.
DO NOT remove any existing features.
DO NOT change the landing page structure.
DO NOT change the animations or responsive design.

Keep the existing design and functionality exactly as it is.
ONLY make the API-related modifications described below.

=====

PROJECT

=====

Project name: CreatorIQ

Purpose: An AI-powered YouTube pre-upload assistant where creators can upload their edited video before publishing and later rec

=====

EXISTING DESIGN -- KEEP EXACTLY THE SAME

=====

Keep the current design:

- Premium dark/black background
- Red YouTube-inspired accent
- Red glowing gradients
- White bold typography
- Dark cards
- Subtle borders
- Floating UI elements
- Modern AI/SaaS appearance
- Responsive layout
- Smooth scrolling
- Floating dashboard
- Red glow effects
- Existing hero section
- Existing "Get Started" button
- Existing Home section
- Existing feature cards
- Existing "How It Works" section

Do NOT redesign anything.

Keep the existing brand name: CreatorIQ

Keep the existing hero: "Make Every Upload Count."

Keep the existing landing page and Home page exactly as currently designed.

=====

DRAG & DROP -- KEEP AND IMPLEMENT

=====

Keep the existing drag-and-drop video upload interface.

Use:

HTML5 Drag and Drop API

HTML5 File API

The user should be able to:

- Drag and drop a video
- Browse and select a video
- Validate the video
- Preview the video
- Display filename
- Display file size
- Display duration
- Display file type
- Remove the selected video

Supported: MP4, MOV, WebM

Keep the existing visual drag-and-drop design.

When dragging:

- Red border
- Red glow
- Smooth transition
- "Drop your video here"

Do not change the existing UI.

=====

ONLY API MODIFICATIONS

=====

Make the frontend API-ready without implementing the actual AI/ML analysis yet.

The APIs should provide the structure needed for future integration.

Use JavaScript Fetch API for communication.

1. VIDEO UPLOAD API: POST /api/upload

- Create FormData, add video file, send via Fetch API.
 - Do not perform AI/ML analysis during upload.
2. VIDEO ANALYSIS API: POST /api/analyze
- Prepare endpoint structure.
 - Add code comment: // AI/ML video analysis will be connected here in the next development phase.
 - Do not calculate fake scores/predictions.
3. ANALYSIS RESULTS API: GET /api/analysis/{id}
- Return video quality, engagement score, views, likes, shares, audience, retention, recommendations.
 - Display placeholders: "AI Analysis Pending", "Will be available after ML integration".
4. YOUTUBE DATA API V3 & 5. YOUTUBE ANALYTICS API
- Create separate service/configuration structure.
 - Data will be used for engagement prediction model training.
 - Never place API key in frontend code; use backend environment variables.
6. FFMPEG / FFPROBE
- Prepare backend architecture for extracting resolution, FPS, duration, bitrate, video codec, audio codec, audio presence.
 - Do not implement quality scoring now.
7. FLASK BACKEND API STRUCTURE
- Endpoints: POST /api/upload, POST /api/analyze, GET /api/analysis/{id}, GET /api/health.
8. MACHINE LEARNING -- FUTURE ONLY
- Do not train models or generate fake predictions. UI placeholders only.
9. EXISTING ANALYSIS DASHBOARD -- KEEP
- Keep cards for: VIDEO QUALITY, ENGAGEMENT SCORE, ESTIMATED VIEWS, ESTIMATED LIKES, ESTIMATED SHARES, AUDIENCE ANALYSIS, VIEWER
 - Display "AI Analysis Pending" or "Coming Soon".
10. API STATUS
- Add subtle API status section to dashboard (Upload API: Ready, Analysis API: Ready for ML integration, YouTube API: Not connected)
11. SECURITY
- Do not expose API keys/secrets in frontend. Use .env in .gitignore.
12. PROJECT STRUCTURE
- ```
creatoriq/
|-- frontend/index.html
|-- backend/ (app.py, routes/, services/)
|-- .env.example, .gitignore, README.md
```